Vanguard (VG) Service Manual

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Author: Dylan Laister

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APPROVALS

VP Engineering	Engineering
Chris Luginbuhl	
Signature:	Date:

Mandatory Reviewers
The undersigned have performed a review of this document

Director of Service	Operations
Rob Murray	
Signature:	Date:

Revision History

Revision	Date	Author	Type (Major/Minor)	Summary of Changes
1.0	20 AUG 2007	Dylan Laister	Major	Initial Release
2.0	18 JAN 2008	Dylan Laister	Minor	Updates for stretcher changes
3.0	4 MAR 2008	Dylan Laister	Major	Rewording for clarity of instructions
4.0	30 JUL 2008	Dylan Laister	Major	Added part and assembly replacement procedures
5.0	12 JAN 2009	Rob Murray	Minor	Grammatical fixes and correction of one acronym
6.0	3 MAR 2009	Dylan Laister	Minor	Updated figures & some edits for clarity

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1 Purpose

This Service Manual is designed to provide detailed instructions to a Service Technician for troubleshooting and repairing the Vanguard (VG) System including the stretcher and RF Coils (mechanical and electrical parts). The Troubleshooting Table below details the required steps to methodically assess and correct the reported problem(s).

Information reported from the site contact explaining the problem, is paramount and the details must be entered into the Problem Management Database. The Service Technician must carefully gather and interpret the information received from the site contact, to ensure all details are enclosed to enable the problem to be quickly diagnosed and repaired.

2 Service Tools Required

Item	Part Number	Qty	Description

1	SMI-0294	1	Service Kit GE Vanguard

3 Definitions and Abbreviations

Throughout this document the following terms and abbreviations will be used:

UBS – Upper Body Support
LBS – Lower Body Support
FSA – Front Support Assembly
RSA – Rear Support Assembly
Head end – Patient Head end of the Vanguard
Foot end – Patient Foot end of the Vanguard
Stretcher – Bottom portion of the Vanguard
Patient support – Top fibreglass portion of the Vanguard
Left/Right –Left/Right directions relative to a patient lying prone (face down) on the system

4 Precautions to be taken

- Before entering the MRI Room with the Vanguard system the following activities must be performed:
- Thoroughly inspect the system to ensure loose ferrous/magnetic items are not contained anywhere on the system. Loose objects can become airborne in the presence of the scanner, potentially causing harm to the Patient, Operator or equipment.
- Do not attempt to make any changes or adjustments to the GE scanner or GE Signa stretcher. The Vanguard is designed to work with the current state of the GE scanner operating in a normal mode
- Patient data is confidential – be sure not to examine, copy or photograph any printed or electronic matter containing patient names, personal details, medical images or other confidential data. (HIPAA Compliancy) Health Insurance Portability and Accountability Act
- ***When replacing any springs on the Vanguard, be sure to do so outside of the scanner room. Some springs in the Vanguard are Stainless Steel, but may be mildly magnetic.***

5 Preparing for the Service Call

Prior to performing the Service Call the following are required:

- Co-ordinate with customer contact and determine date of service visit.
- A copy of the P.O. for the unit to be serviced.
- Contact information for the site (usually the charge tech or administrator) and back-up contact person
- Review any pending actions from the Problem Management database specific to the site
- Determine if any nearby sites can be visited during the same trip to perform preventative maintenance
- Bring a complete service kit, camera, notebook, service visit cards and personal business cards
- When international travel is required, a signed letter from the service manager or other manager (“Service Rep Letter”) explaining the purpose of the trip, the destination, dates and the employment status of the repair technician
- ID badge, to be kept on person at all times, and passport/visa (as needed).
- Hotel info, customer site address, directions to/from site, airport and hotel. If possible, contact the site specific GE Service Personnel (to inform them of the pending service call and request information on any recent problems or service with the scanner (service calls can be generated from recent service to the scanner involving adjustment of mechanical parts or software upgrades and patches)
- Ensure that only Sentinelle Service Personnel make adjustments or repairs to the system
- Review Sentinelle service logs for the site for previous trip reports and service pending.

6 Performing and Recording the Service Call

Record the results of all activities performed including:

- Record *all activities undertaken* and the results, functionality checks of all safety mechanisms and interlocks, and the condition of the scanner
- Record any comments or concerns received from site
- Leave a completed service card and inform relevant staff members who are present that service is complete and indicate state of functionality. If the system fails any of the functionality tests below, it must be taken out of service.

7 Troubleshooting Guide for Mechanical Components

7.1 Troubleshooting Methodology

The following steps will guide you to identify and resolve the problem

1. ***Identify*** the malfunction.

2. Review the ***possible causes*** of the malfunction, considering User error and non-technical issues which may be impeding function of the Vanguard.

3. Execute the ***actions*** which correspond to the possible causes.

4. Damaged parts may be the cause of the reported problem. ***Repair or replace*** affected parts or hardware and then complete any associated adjustments

**Note: Every action includes a final "verification" step, including all interlocks and sub-assemblies. This full system check is performed for preventative maintenance.*

7.2 Potential causes Triggering Service Call

- Damage caused during use - to parts and assemblies
- User error in operating the system
- Incorrect interpretation of the problem being reported
- Improper training/Improper use of system
- GE service related changes to the scanner (e.g. installed/upgraded software patches)

7.3 Table 1 – Mechanical Troubleshooting Guidance

Table 1 details potential problems, causes, checklist items and possible actions to resolve problem

PROBLEM	POSSIBLE CAUSES	CHECKS to be Performed	REPAIR or REPLACE
Unable to dock stretcher to scanner	• Items interfering with docking • Docking hook misaligned • Docking pedal jammed • Damage to docking mechanism. • Stretcher-scanner gap not to specification • Caster(s) locked	• Investigate damage or misalignment to docking mechanism and • Foreign objects present • Review Procedures: ✓ 7.4 Alignment ✓ 7.7 Bumper Adjustment ✓ 7.8 Gap Adjustment ✓ 7.9 Docking Force Adj. ✓ 7.10 Bumper Plunger Adj. ✓ 7.16 Final Verification	➢ 9.10 Swing Assembly ➢ 9.2 Spring Bumper Assembly ➢ 10.13 L-R Centering Bumper ➢ 9.11 Docking Linkage Assembly ➢ 10.2 Tension Coupling ➢ 10.14 Caster
Unable to undock stretcher from scanner	• Damage to docking mechanism • Undock pedal jammed • Patient support not in home position • Caster(s) locked	• Check bridge is closed • Check for damage/misalignment to parts of docking mechanism and foreign objects. • Review Procedures: ✓ 7.4 Alignment ✓ 7.7 Bumper Adjustment ✓ 7.8 Gap Adjustment ✓ 7.9 Docking Force Adj. ✓ 7.10 Bumper Plunger Adj. ✓ 7.12 No-undock Hook Adj. ✓ 7.14 Doghouse Mech. Adj. ✓ 7.16 Final Verification	➢ 9.9 Bridge Latch Assembly ➢ 9.25 Latch tab and spring ➢ 10.15 Cable elbows and cable ➢ 9.20 Doghouse Mechanism ➢ 9.10 Swing Assembly ➢ 9.2 Spring Bumper Assembly ➢ 10.13 L-R Centering Bumper ➢ 9.11 Docking Linkage Assembly ➢ 10.12 Tension Coupling ➢ 10.14 Caster
Patient support will not advance from stretcher and/or in scanner	• Bridge not closed • Stretcher not docked • Damage to patient • support hardware • Items impeding motion of patient support • Doghouse interlock mechanism not actuated • Scanner's bridge higher/lower • Latch tab protruding • Stretcher not centred	• Check for damage/misalignment to parts of patient support hardware and foreign objects. • Review Procedure: ✓ 7.8 Gap Adjustment ✓ 7.13 Centering Guide Adj. ✓ 7.14 Doghouse Mech. Adj. ✓ 7.15 Latch Tab Adjustment ✓ 7.16 Final Verification	➢ 9.9 Bridge Latch Assembly ➢ 9.25Latch tab and spring ➢ 10.15 Cable elbows and cable ➢ 9.20 Doghouse Mechanism ➢ 9.10 Swing Assembly ➢ 9.2 Spring Bumper Assembly ➢ 10.13 L-R Centering Bumper ➢ 9.11 Docking Linkage Assembly ➢ 10.14 Caster
Patient support stuck in scanner or on stretcher and cannot advance or return to home position	• Bridge not closed • Items impeding motion of patient support. • Latch tab protruding • Docking sensor not actuated • Stretcher-scanner gap not in spec. • Stretcher not centred	• Check for damage/misalignment to parts of doghouse mechanism, docking mechanism and patient support hardware. • Review Procedures: ✓ 7.7 Bumper Adjustment ✓ 7.9 Docking Force Adj. ✓ 7.10 Bumper Plunger Adj. ✓ 7.13Centering Guide Adj. ✓ 3.14 Doghouse Mech. Adj. ✓ 7.15 Latch Tab Adjustment ✓ 7.16 Final Verification	➢ 9.10 Swing Assembly ➢ 9.2 Spring Bumper Assembly ➢ 10.13 L-R Centering Bumper ➢ 9.11 Docking Linkage Assembly ➢ 10.12Tension Coupling Docking Linkage ➢ 9.9 Bridge Latch Assembly ➢ 9.25Latch tab and spring ➢ 10.15 Cable elbows and cable

Patient support comes unlatched from doghouse when advancing or returning	• Items preventing latching of doghouse to patient support. • Emergency release being actuated • Stretcher not centred • Stretcher not docked straight • Stretcher not level	• Check for damage/misalignment to parts of doghouse mechanism, docking mechanism and patient support hardware. • Review Procedures: ✓ 7.4 Fine Height Adj. ✓ 7.7 Bumper Adjustment ✓ 7.9 Docking Force Adj. ✓ 7.10 Bumper Plunger Adj. ✓ 7.13 Centering Guide Adj. ✓ 7.14 Doghouse Mech. Adj. ✓ 7.15 Latch Tab Adjustment ✓ 7.16 Final Verification	➢ 9.10 Swing Assembly ➢ 9.2 Spring Bumper Assembly ➢ 10.13 L-R Centering Bumper ➢ 9.9 Bridge Latch Assembly ➢ 9.25 Latch tab and spring ➢ 10.15 Cable elbows and cable
Rough or noisy motion of patient support in scanner or on stretcher	• Bridge not closed • Latch tab protruding • Doghouse mechanism not fully actuated • Items impeding motion of patient support • Damage to wheels and tabletop hardware	• Check for damage/misalignment to parts of patient support hardware and foreign objects. • Review Procedure: ✓ 7.4 Alignment ✓ 7.6 Fine Height Adj. ✓ 7.7 Bumper Adjustment ✓ 7.16 Final Verification	➢ 9.9 Bridge Latch Assembly ➢ 9.25 Latch tab and spring ➢ 10.15 Cable elbows and cable ➢ 9.2 Spring Bumper Assembly ➢ 10.13 L-R Centering Bumper ➢ 10.12 Patient Support Wheel

7.4 Alignment

Alignment problems can affect the proper functioning of the Vanguard System. If the service problem affected the scanner's bridge or dock, you may be required to correct the alignment.

1. Remove the patient support from the stretcher and place in a secure location. Ensure that the patient support is not scratched, scuffed or damaged while doing so.

2. Remove the drawers from the stretcher by depressing the 2 plastic latches on either side and pulling the drawers out of the rear support assembly. Place them away from the work area. (Fig 1)

Fig 1

3. Screw in the doghouse sensor actuator in the middle of the bumper as far as possible turning it in a clockwise direction by hand.

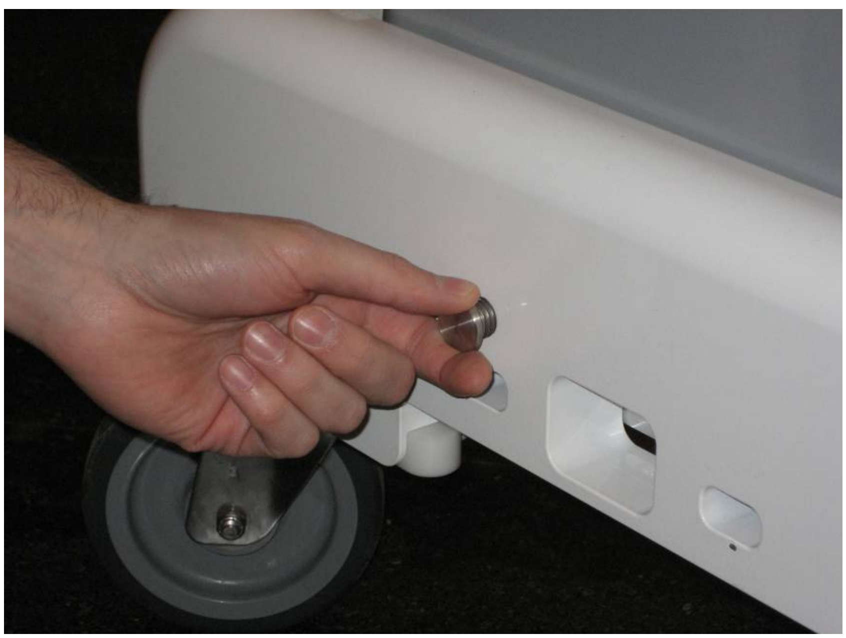


Fig 2

4. Adjust the docking hook height by loosening the socket head cap screw with a 3/16” Allen key. Adjust the position of the round eccentrically mounted hook stop to allow the hook to enter the docking fixture. When properly positioned, the hook is above the bronze docking hook and is not contacting the inside top surface of the GE docking fixture. **(Fig 3)**

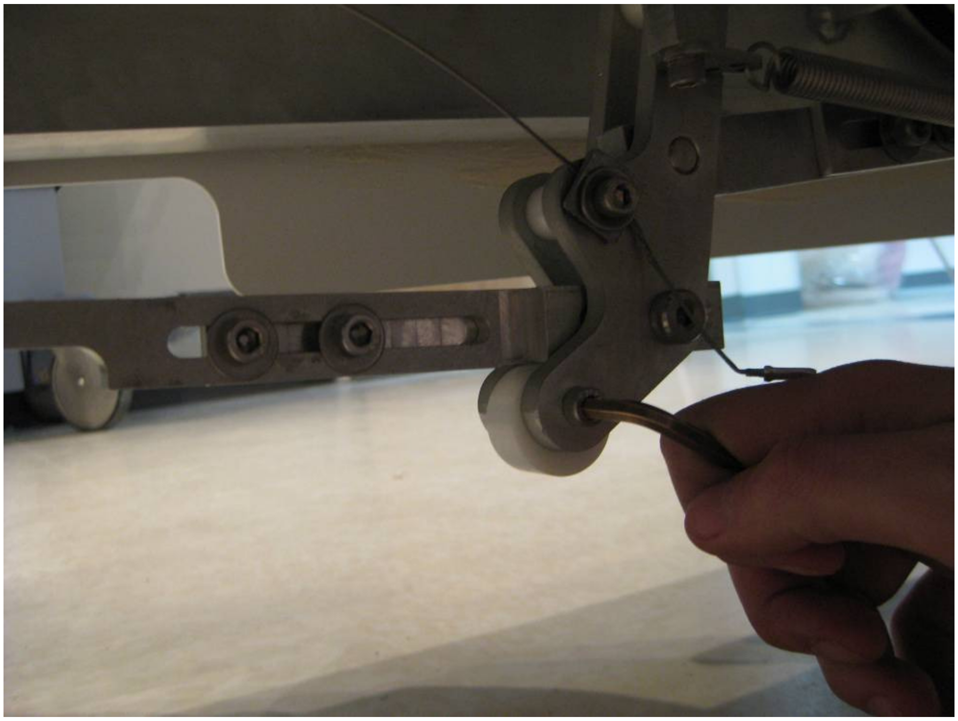


Fig 3

5. Align the Vanguard system to the scanner carefully, ensuring that the bumper does not contact the fixed static pin on the docking fixture. The static pin should be aligned with the corresponding clearance pocket in the bumper. Measure the height difference of the stretcher and the Scanner's bridge with the MRI compatible ruler. *(Fig 4)*

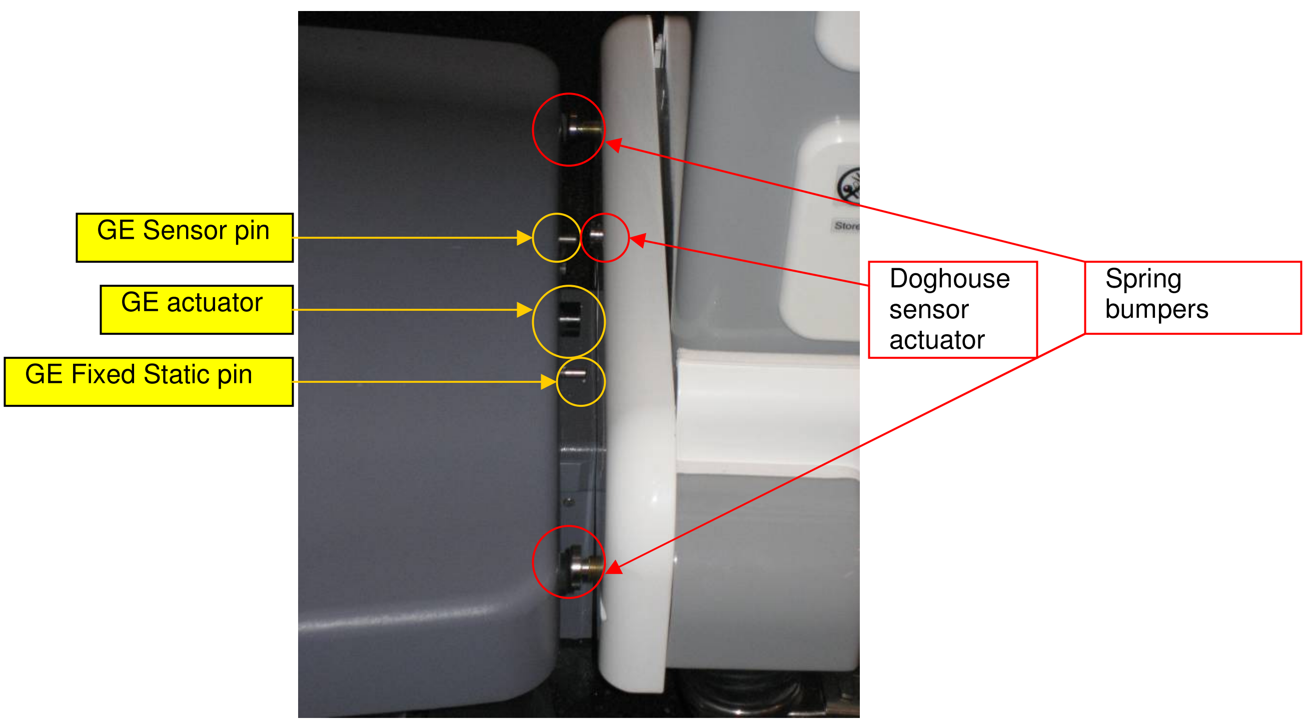


Fig 4

6. If the height difference between the stretcher and the Scanner's bridge is greater than 1/2", perform course height adjustment, followed by fine height adjustment. Coarse adjustment can only be

performed in increments of 1/2". If the height difference is less than 1/2" then fine adjustment may be required.

7.5 Coarse Height Adjustment

Coarse height adjustment is performed when the height difference between the Scanner's bridge and the Vanguard LBS is 1/2" or greater.

1. Loosen the cable by undoing the round head screw actuating the no-undock hook with a 5/32 Allen key as shown below. **(Fig 5)**

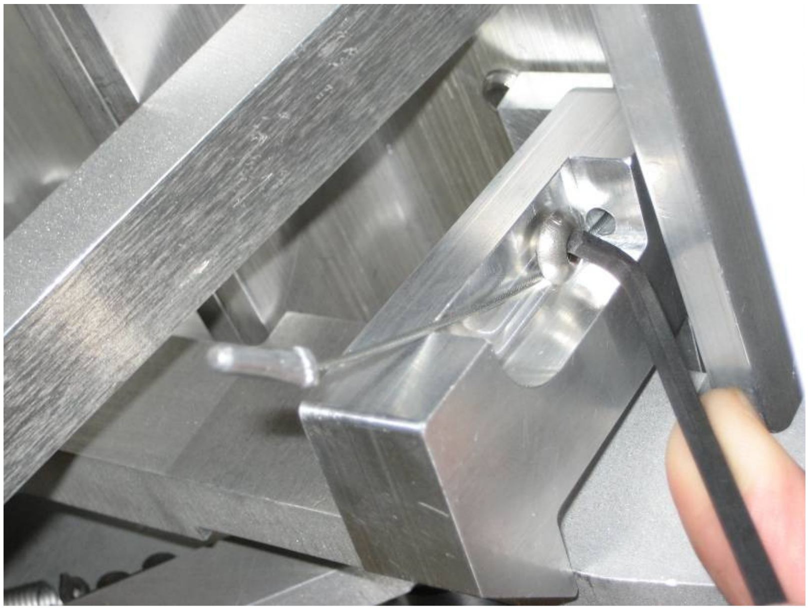


Fig 5

2. Using a 1/8" Allen Key remove the LBS skirt by unscrewing the two external mounting round head screws and loosening the third screw. Slide the skirt down to the bottom of the RSA and separate it by bending it slightly. Be careful not to scratch any of the paint. (Fig 6)

Fig 6

3. Using a 5/32" Allen key remove the two half skirts from the FSA of the stretcher by unscrewing the eight round head screws (Fig 7)

Fig 7

4. Open the battery compartment access cover on the FSA by unscrewing the thumb fastener (Fig 8) and then pivoting the assembly into its open position. (Fig 8.1)

Fig 8

Fig 8.1

5. Take note of the position of the 4 bolts in the series of vertical holes inside the FSA and RSA. (Fig 9)

Fig 9

6. Using a ½" wrench, adjust the height of the UBS by removing 4 nuts and lock washers. Remove three of the four carriage bolts.

Caution: do not remove all four of the carriage bolts at the same time.

7. Adjust by tilting the UBS to the left and right and inserting and removing the bolts as necessary.

When the correct height has been achieved, re-install the four sets of nuts, lock washers and bolts. (Fig 9.1)

Fig 9.1

8. Adjust the height of the UBS one side at a time by tilting it in the head/foot end direction and re-install the 4 sets of nuts, lock washers and bolts.

Upon completion of coarse height adjustment, verify the height adjustment and then replace the skirts on the UBS and LBS.

7.6 Fine Height Adjustment

Fine height adjustment is performed when height adjustment of less than 1/2" is required or after coarse height adjustment has been performed.

1. Align the stretcher to the scanner, ensuring the bumper does not contact the fixed static pin on the docking fixture, and record the height differential.

2. Using a 5/32” Allen Key remove the four wheel covers from the stretcher by loosening eight round head screws and sliding the covers off of the wheels. (Fig 10)

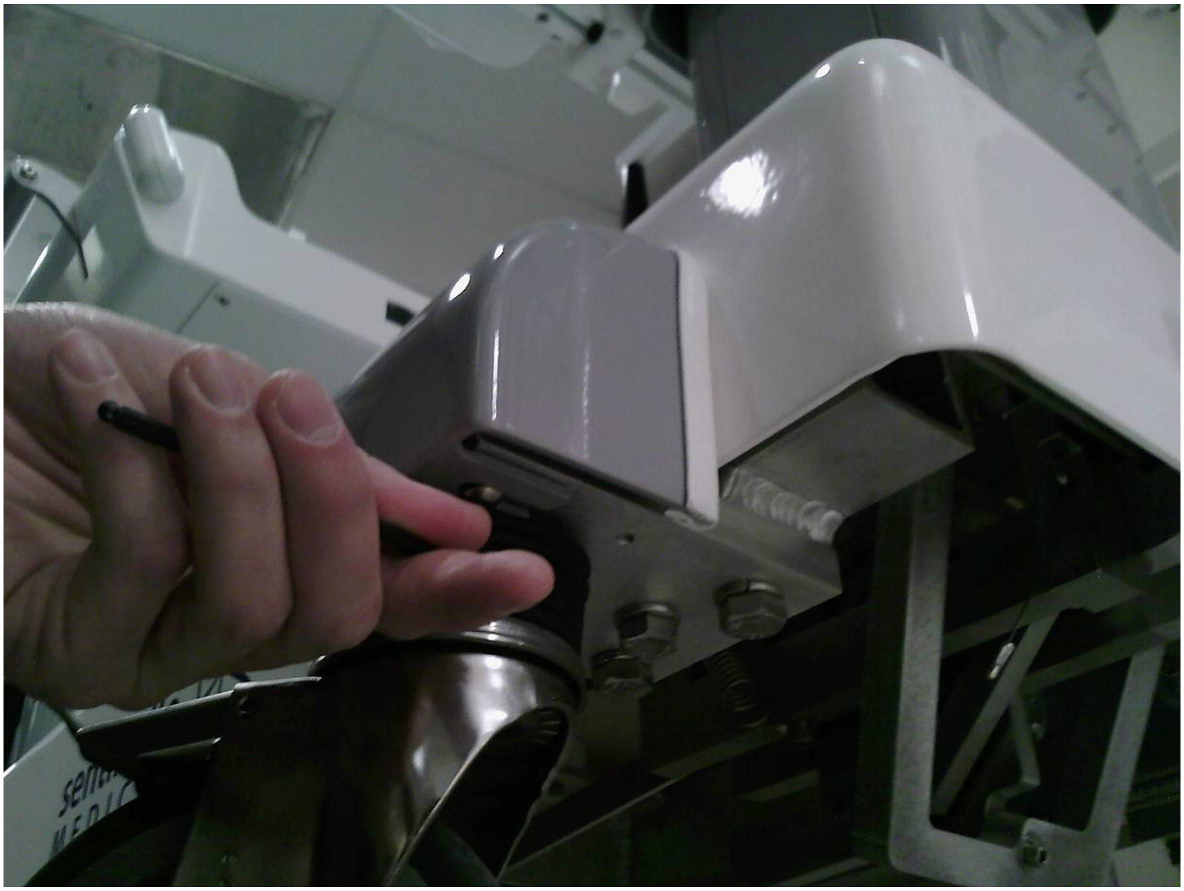


Fig 10

3. Loosen each of the large nuts at least four turns using the 1.5” MRI compatible wrench. (Fig 11)

Fig 11

4. Adjust the four caster height by adjusting hex bolts using the 1.5” MRI compatible wrench (CW from top to raise the stretcher, CCW to lower the stretcher.) Ensure that all four hex bolts are initially adjusted by the same number of turns (**Note that 1 turn = 1/8” of height adjustment**).

5. It is more important to align the stretcher with the scanner’s bridge than to ensure it is level or parallel to the floor. Use a straightedge to check the co-planarity of the stretcher and the scanner’s bridge. Lay it across the gap between the stretcher and the scanner’s bridge in the regions where the wheels run. Assess the continuity of the two surfaces. Note that the scanner’s bridge has a slight bevel or slope at its edge. The lower body support must be within 1.5mm (1/16”) of the main section of the bore, disregarding the sloped region. (Fig 12)

Fig 12

Note: The floor of the MRI room may be uneven, which must be adjusted for. The stretcher height at both the head and foot end must match the scanner floor height. This can be gauged by using a straightedge and visually inspecting the stretcher from the head end. Adjust the head and foot end casters to eliminate the slope and re-evaluate

5. Verify all wheels contact the floor when the stretcher is in the docked position. This is checked by manually spinning the casters to determine if stretcher weight is present. Adjust any non-contacting casters to optimize the level of the stretcher.

6. Using the 1.5" MRI compatible wrench, tighten each of the large nuts very tight. It is necessary to brace against the stretcher and use both hands for high torque.

7. Replace the wheel covers on the stretcher and tighten the round head screws using a 5/32" Allen key. Ensure the covers fit tightly to the shroud.

7.7 Bumper Plate Adjustment

The bumper plate which interfaces with the GE docking fixture must be centered to the GE docking fixture to permit function of the Vanguard.

1. With the stretcher undocked, loosen the two ¼"-20NC bolts on the hidden face of the bumper which lock the LR adjustment bumpers in place with a 3/16" Allen key (Fig 13)

Fig 13

2. Loosen the two left-right fine adjustment bumpers which are located in slots on the bottom of the bumper with a ¼" Allen key and slide them out so they protrude to their fullest extent. (Fig 14)

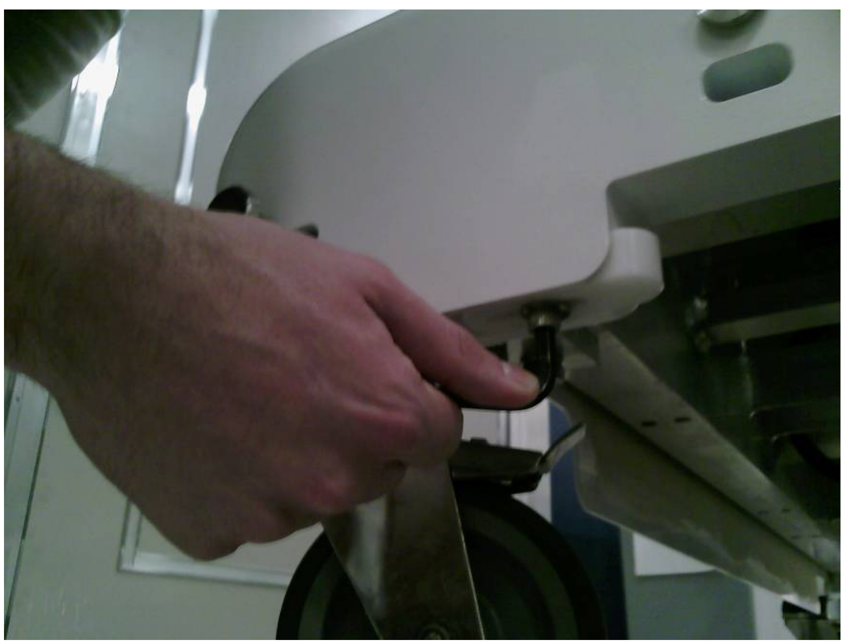


Fig 14

3. Dock the stretcher to the scanner, while manually positioning it such that the guide rails of the stretcher and of the Scanner's bridge are collinear. (Fig 15)

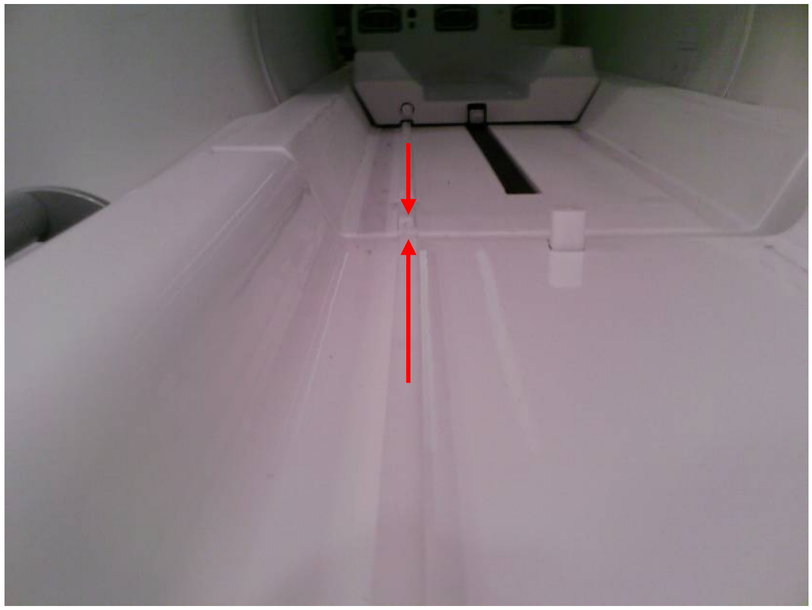


Fig 15

4. The bumper must be centred to the docking fixture with respect to the corresponding rubber features for the spring bumpers, static pin and sensor pin. Note the amount that it must be moved in order to be centred (Fig 16)

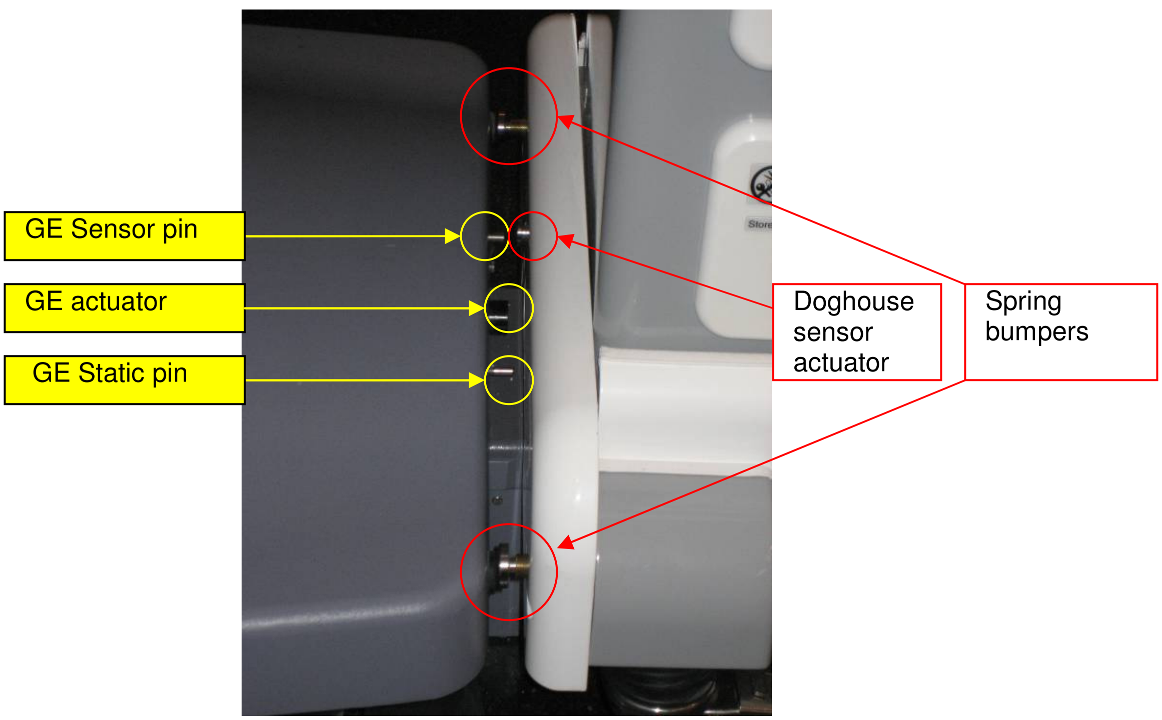


Fig 16

5. Undock the stretcher. Loosen the four socket head cap screws with a 1/4” Allen key. Adjust the position of the bumper plate L-R and A-P (up - down) as required; ensuring it is vertically and horizontally centred to the docking fixture. Dock the stretcher and check the centering of the bumper. Undock and adjust and reposition if required. (Fig 17)

Fig 17

After the bumper is centred, tighten the four socket head cap screws to secure the bumper plate to the stretcher using a 1/4” Allen key. Ensure it does not shift while being tightened.

7.8 Stretcher to Scanner Gap Adjustment

The gap adjustment is required to eliminate a possible pinch point between the Vanguard LBS and the Scanner’s bridge.

1. Dock the stretcher to the scanner and measure the gap between the stretcher LBS and the scanner with a MRI compatible ruler. This gap must be 1” (+1/16”/ -0) when the system is docked. (Fig 19)

Fig 19

2. If adjustment is required, move the lower body support by loosening with a 9/16" socket or ratchet, the four hex head bolts located inside the LBS which fasten it to the RSA. Slide the LBS accordingly to leave a 1" gap. Tighten the four bolts firmly to fasten the LBS in place. (Fig 20)

Fig 20

7.9 Docking Force Adjustment

Setting the docking force requires incremental adjustments of the docking hook until adequate force is achieved. If the docking force is too high it will result in premature wear to the Vanguard.

1. With the stretcher still positioned against the GE docking fixture, loosen the two socket head cap screws on the docking hook assembly using a 3/16” Allen key and adjust the hook by sliding it away from the docking fixture until it makes contact with the bronze hook in the docking fixture.

2. While applying pressure from the head end of the stretcher to compress the bumper springs, depress the docking pedal and release it slowly to prevent any jarring motion to the docking hook.

3. Adjust the hook by sliding it 1/8” away from the docking fixture and tighten one of the two socket head cap screws using a 3/16” Allen key.

4. Dock the stretcher by depressing the docking pedal and take note of the force required to dock using an MR-safe spring scale. The force must be between 30 and 40 lbs.

5. Test the docking force adjustment while docked by attempting to pull the stretcher away from the scanner and pushing it towards the scanner with some force. Test the docking force by pushing the stretcher left and right from the head end with light to moderate force. Also verify that the bumper spring plungers remain bottomed out despite side loading. If the stretcher comes undocked or

doesn't remain firmly docked to the scanner with the bumper spring plungers bottomed out, adjust the docking hook tighter in 1/16" increments while repeating the aforementioned testing as required.

6. When appropriate docking force is achieved, firmly tighten the two socket head cap screws on the docking hook assembly using a 3/16" Allen key.

7.10 Bumper Plunger Adjustment

Similar to centering adjustment, this adjustment will ensure smooth travel of the patient support by making sure the centering features of the Scanner's bridge and Vanguard are collinear.

1. Dock the stretcher to the scanner and take note of the position of the left and right spring bumpers and the co linearity of the stretcher and Scanner's bridge guide rails. When the system is docked the spring bumpers must be fully depressed.

2. If the guide rails are not collinear, undock the system and adjust the brass spring bumper housings accordingly. Always tighten one plunger while loosening the other plunger the same amount, so that the docking force is not affected. Dock the system and check the ridges again for co linearity. Repeat adjusting the spring bumper housings as required. (Fig 18) Ensure the stretcher-scanner gap and the docking force is still within specification.

Fig 18

3. Adjust the height of the center bumper until it depresses the plunger on the docking fixture by between 1/8" and 3/16" when the stretcher is positioned against the scanner.
4. Use a feeler gauge to ensure the plunger is not bottomed out or overloaded.

7.11 Head End Position Adjustment

This adjustment is made in relation to the gap between the stretcher and scanner. If the LBS is moved, the UBS must also move the same amount in the same direction. The UBS determines the home position of the patient support.

1. Prepare to reposition the UBS by loosening the four hex head bolts on the underside of the UBS using a 9/16” socket or wrench. Push the UBS toward the foot end so the travel stops of the UBS are firmly against the patient support. (Fig 24)

Fig 24

2. Close the bridge such that the spring plungers are just contacting the LBS and the square hole in to which the bridge mechanism is inserted is visible. (Fig 25)

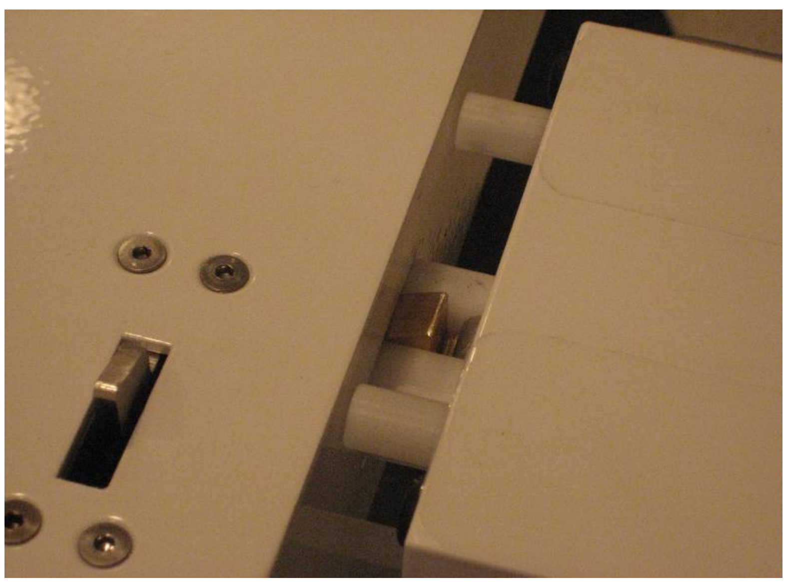


Fig 25

3. Insert a 1/32” spacer such as the MRI compatible stainless steel ruler between the latch tab and the foot end of the latch fixture and adjust the position of the UBS by firmly pushing the patient support towards the head end of the stretcher. (Fig 26)

Fig 26

4. Ensure that the bridge mechanism is entering the square hole without interference and make any adjustments that may be required.

5. Fasten the UBS in place by tightening the four hex head bolts using a 9/16” wrench.

6. Remove the ruler from between the latch tab and the patient support and check to verify that the latch tab moves without rubbing any part of the latch fixture by depressing and releasing the docking pedal or depressing the tab by hand.

7.12 No-undock Hook Adjustment

The no-undock hook is an interlock which prevents the stretcher from being undocked when the patient support is not in the home position.

1. Dock the system and advance the Patient support roughly one foot into the scanner. While the Patient support is advanced and the no-undock hook is in the down position against its travel stop, pull the cable firmly downward and clamp it to the no-undock hook by tightening the round head screw with a 5/32” Allen key. (Fig 27)

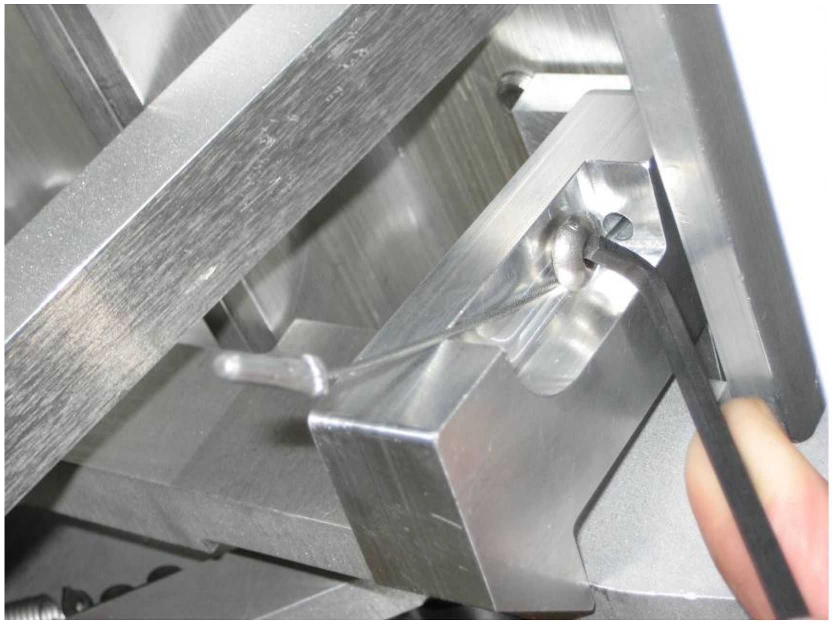


Fig 27

2. Return the Patient support to the home position and check the no-undock hook to verify it is functioning correctly (being at least 1/8” clear of the undock pedal arm).

Note: The entire hook mechanism can be moved L-R if the hook is touching its travel stop but is not at least 1/8” clear from the undock pedal arm.

7.13 Centering Guide Adjustment

Proper centering of the Vanguard will ensure that the patient support will transit to the Scanner’s bridge smoothly.

1. Dock the stretcher by slowly depressing the dock pedal while carefully aligning the guide rails of the stretcher and the Scanner’s bridge. This may take more than one attempt. Use a straightedge to verify the co-linearity of the ridges. Slide the two left-right fine adjustment bumpers so they firmly contact the scanner’s docking fixture and fasten them firmly in place by tightening the socket head cap screws with a ¼” Allen key.

2. Undock, reposition the stretcher and dock the stretcher again to ensure that it is centered. Re-adjust the left-right bumpers if required and check again. When the stretcher is pressed up against the scanner but undocked, there should be no more than 1/16” of L-R play. Once the stretcher is centered, undock the stretcher and firmly tighten the two socket head cap screws on the back side of the bumper plate with a 3/16” Allen key.
3.

7.14 Doghouse Mechanism Adjustment

The doghouse mechanism is a secondary interlock that prevents the patient support from being rolled off the stretcher when not docked to the MRI.

1. Dock the stretcher to the scanner.

2. As the doghouse advances to latch with the patient support, ensure the scanner’s doghouse plunger fully depresses the doghouse mechanism screw on the Vanguard patient support. If it does not, take note of the distance required for it to do so (Fig 28)

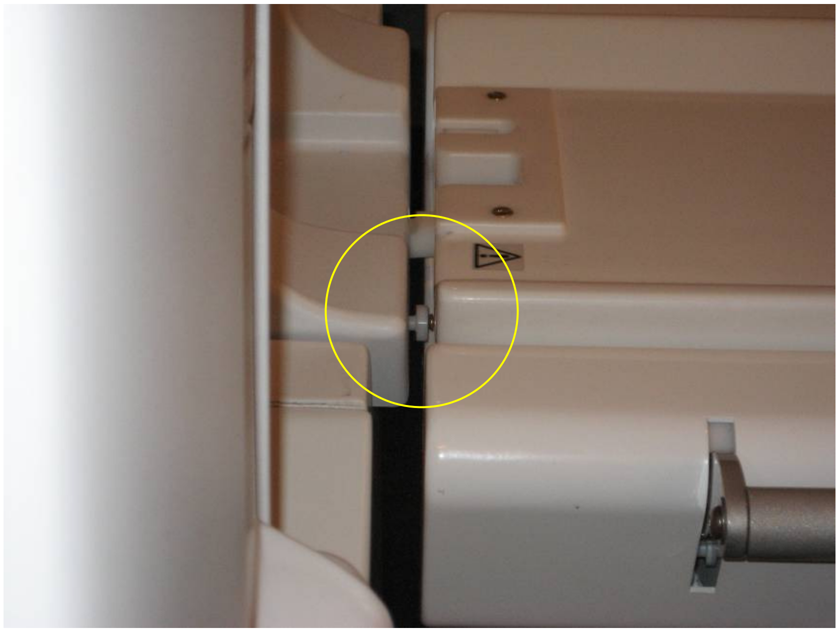


Fig 28

3. If adjustment of the doghouse mechanism is required, unlatch the doghouse manually, holding up the doghouse latch hook while advancing the doghouse into the scanner. First loosen off the locking nut and then with a Phillips screwdriver, unscrew the brass adjusting screw the required amount and tighten it with the nut.

4. Ensure that the plunger is not overloaded at any point by proceeding with adjustments in small increments of 1/16" and checking that a feeler gauge may be freely inserted and removed from between the plunger and adjuster at any point.

5. Advance the doghouse back to the patient support. Ensure that the patient support can advance freely and the doghouse mechanism is not rubbing against the guide rails of the Patient support and the scanner's bridge. Once the desired adjustment has been achieved, tighten up the locking nut on the brass adjusting screw. Be sure to use a wrench to avoid gradual loosening.

7.15 Latch Tab Adjustment

1. Dock the stretcher to the scanner.

Fig 29

2. Remove all slack from the cable by pulling it firmly from where is terminates after passing under the cable clamp and re-tighten the socket head cap screw securing the cable clamp using a 5/32" Allen key. (Fig 30)

3. Undock and dock the Vanguard. Ensure that the latch tab is no more than 1/32" above the stretcher surface and no more than 1/16" below when the dock pedal is depressed. If latch tab is still protruding when the Vangaurd is docked, undock the Vanguard, loosen the cable clamp and advance the cable as required until the latch tab retracts fully when the dock pedal is at the end of its stroke. .Ensure that the latch tab does not bottom out before the pedal stroke is complete – this will overload the cable and bend the cable elbows.

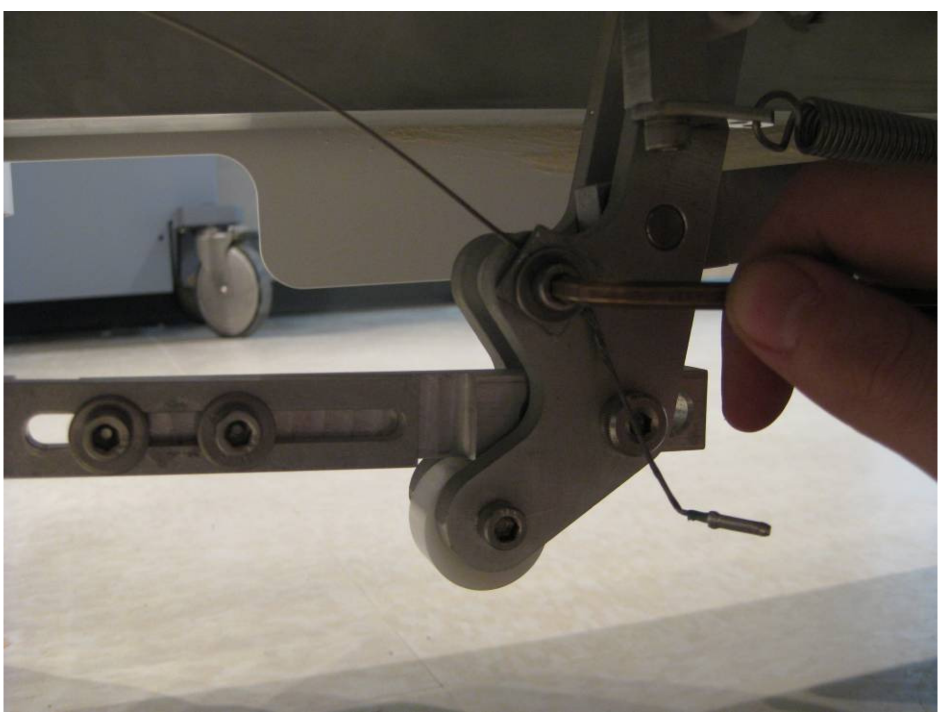


Fig 30

7.16 Final Verification and checks of Mechanical Components/Assemblies

Final verification of install is undertaken to ensure that the installation procedure was executed correctly. If a malfunction is detected during any of the verifications, review the pertinent procedure and test again. If a malfunction is detected due to damaged parts, the part must be replaced and the verification performed again.

1. Replace the drawers if they were removed. Test them to ensure they open and close smoothly.

2. Verify the function of the bridge latch and locking mechanism by opening and closing the bridge with the patient support in the home position. The tabletop must be prevented from advancing to scan using the MRI's controls when the bridge is open.

3. Verify the proper function of the travel stop plunger mechanism by advancing the patient support into the scanner and depressing the travel stop plunger by hand. It should not show any signs of friction. Verify the travel stop plunger permits the stretcher to undock only when the patient support is at the home position.

4. Verify the no-undock hook is lifted at least 1/4" clear of the undock pedal arm when the patient support is in the home position. The hook must also touch its travel stops at either end of its range of motion.

5. Verify the correct function of the latch tab mechanism by stepping on the dock pedal. The latch tab must retract flush or up to 1/32" above or 1/16" below the top surface of the LBS when the docking pedal is pressed fully and should not rub against any part of the patient support when it is being actuated.

6. Verify the correct function of the doghouse interlock. It must move freely and fully actuate the doghouse mechanism when the doghouse is latched to the patient support without overloading the white plunger on the doghouse.

7. With the stretcher away from the MRI, close the bridge and depress and hold the docking pedal completely. It should not be possible to roll the tabletop off the stretcher with 30lbs (14kg) of force.

8. Ensure the catchment opens fully, does not interfere with any part of the Vanguard and that it closes and stays latched when patient support moves in and out of the scanner.

9. Verify the correct function of the side handles. They must actuate fully between locked positions and the white plastic covers must not leave any gaps larger than 5/16" (8mm)

10. Verify the correct function of the Vanguard docking and undocking. When docked, the Vanguard should not come undocked when it is subject to force in any direction. Apply 50lbs of force to the stretcher handle toward the scanner to ensure the Vanguard stays docked.

11. Verify the correct function of the patient support advancing to scan and returning to home. While advancing to scan apply enough resistance to the patient support to cause the doghouse to stop and continue to advance to scan and then return to home position. The patient support should not come unlatched or produce any noise or signs of friction with the scanner bridge or bore while advancing to scan with body weight applied to it.

12. Verify the function of the emergency release by advancing the tabletop to the scan position and then actuating the release to free the table. It should release when the handle is twisted 90 to 110 degrees and the patient support is pulled with minimal force. The patient support should latch securely to the doghouse again after the emergency release handle is released gently.

13. Check that all caster roll and swivel lock brakes work properly.

14. Ensure that a complete set of pads is present, notify Sentinelle of any pads that appear to be damaged or missing.

15. Verify correct function of all four lights on each side of the stretcher. Ensure there are no non-functioning LEDs

16. Ensure all three bumper plungers are securely loc-tited. If not, apply wicking Loctite 290 to the two bumper spring housings and doghouse sensor plunger where they screw in to the bumper plate and quickly wipe off the excess from the bumper and bumper spring housings to prevent staining.

17. Review all technical notes, service bulletins and additional procedures related to the Vanguard for GE Signa MRIs and ensure all actions are complete.

8.0 Electrical Trouble Shooting Guidance

The following section details basic electrical troubleshooting. All other problems should be referred to Sentinelle's engineering support team for more advanced troubleshooting.

PROBLEM	POSSIBLE CAUSES	CHECKS to be Performed	REPAIR or REPLACE
1. Poor Image Quality (Signal Drops, Noise; Artefacts, low SNR etc.)	• Coil Configurations are incorrectly entered or detected after upgrade • Discrete Pins for preamp inside HYP plug are bent • Coil or Cable Tray Failure (pre-amps or cables)	• Run Manual Prescan and check for number of receivers (2, 4, or 8) • Check that pins are not damaged inside plug • Run Manual Prescan to check peaks on all receivers. (flat or no peak indicate failures)	Re-enter/Repair Coil Configurations 9.23 Replace Cable tray Replace Coil(s) or 9.23 Cable Tray
2. Pre-Scan Failure Error: "Open Circuit"	• Coil not properly connected to cable tray • (HYP) Cable Plug not properly connected to scanner • Discrete and/or Coax Pins inside plug are bent • Intermittent/broken electrical connection	• Ensure coil is properly connected • Ensure HYP Cable connector is fully seated and the Green Light is ON • Center pin of coax and discrete pins are not damaged inside plug • Check all cables for electrical continuity	Connect coil Connect Cable Plug 9.23 Cable Tray Replace coil or cable
3. Pre-Scan Failure Error: "Signal too Small"	• Discrete Pins for preamp inside HYP plug are bent • (HYP) Cable Plug not properly connected to scanner- Valid coil set not connected to system • Medial Terminator Plug is not plugged into system for interventional procedures only	• Pins are not damaged inside plug • Ensure HYP cable Plug is fully seated and the Green Light is ON • Check physical coil connection (HYP Cable is connected and green light is present) • Center pin of coax and discrete pins are not damaged inside plug	Replace Cable Tray Connect to Scanner Attach appropriate coil array 9.23 Replace Cable Tray

	• Discrete pins are bent/damaged on Coils or HYP Cable		
4. Vanguard is not recognized by the scanner (Green Light is not present)	• Not all coils are properly connected to cable tray • Medial Terminator Plug is not plugged into cable tray for interventional procedures only • Valid coil set not connected to system • Discrete pins are bent/damaged on Coils or HYP Cable Plug	• Ensure HYP Cable Plug is fully seated and the Green Light is ON • Check physical coil connection to the HYP Cable Tray • Ensure the Medial Terminator Plug is connected to HYP cable tray • Ensure valid coil sets are connected to cable tray • Ensure discrete pins are not damaged inside HYP Plug or other side of Cable Tray where coils are connected • Ensure discrete pins are not damaged inside coil connectors	Repair Connection Connect Coils properly Connect Medial Plug 9.23 Replace Cable tray Replace Coil(s) or Cable Tray
5. Lighting system is inoperative	• Loose connectors • Damaged cables • Dead batteries • Blown fuse	• Check all connectors • Check all cables for damage • Check batteries • Check fuse	• Replace batteries • Replace fuse • 9.5 Replace Wiring Harness

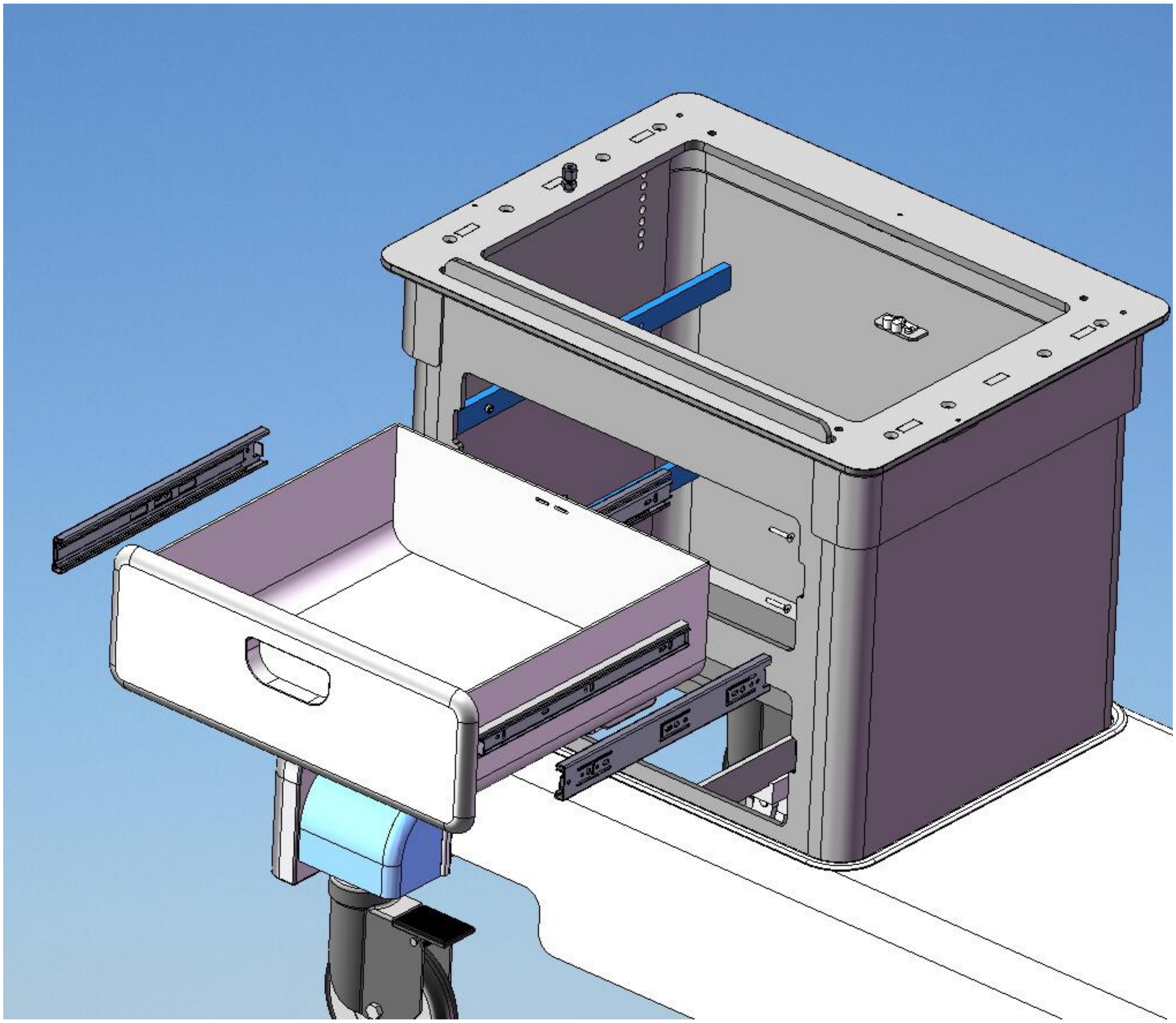
8.1 Final Verification and checks of Electrical Components/Assemblies

After the repair has been completed on any/all electrical components or assemblies (RF Components) perform a Quality Assurance Check as detailed in User Manual – Chapter 3. Use Appendix 1 to record the values obtained and return this documentation to appropriate Sentinelle Medical employees.

9.0 Replacing Assemblies

9.1 Drawer assembly

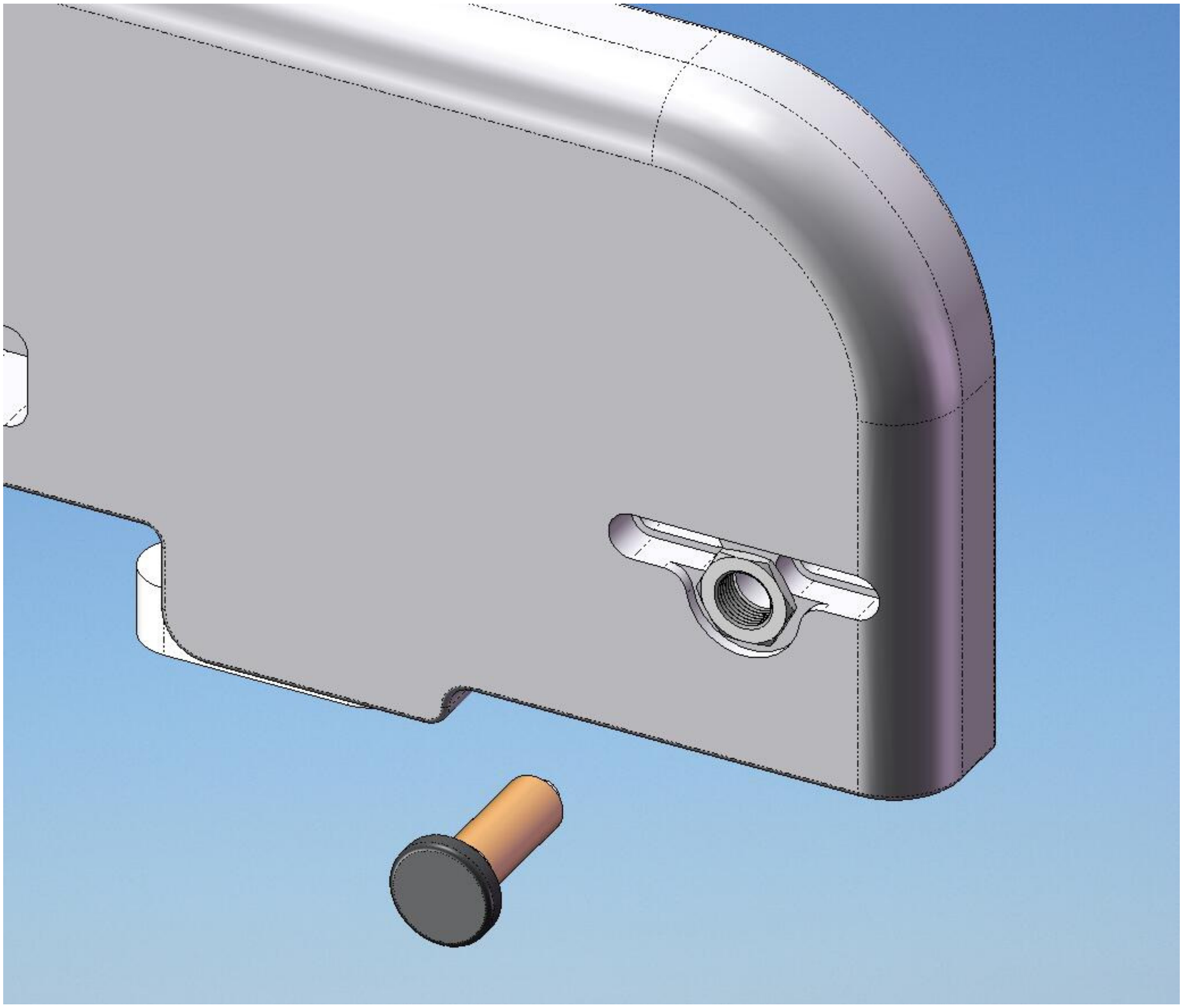
Time Required: 40 minutes
Tools Required: 1/8" on-magnetic Allen wrench
Parts Required: BEP-1000-DWR-000-000



1. Remove drawer from RSA
2. Remove drawer sliders from RSA
3. Fasten new drawer sliders to RSA
4. Insert new drawer to RSA
5. Test drawer for smooth motion.

9.2 Bumper Spring Plunger Assembly

Time Required: 50 minutes
Tools Required: Non-magnetic adjustable wrench
Parts Required: BEP-1000-BSA-000-000

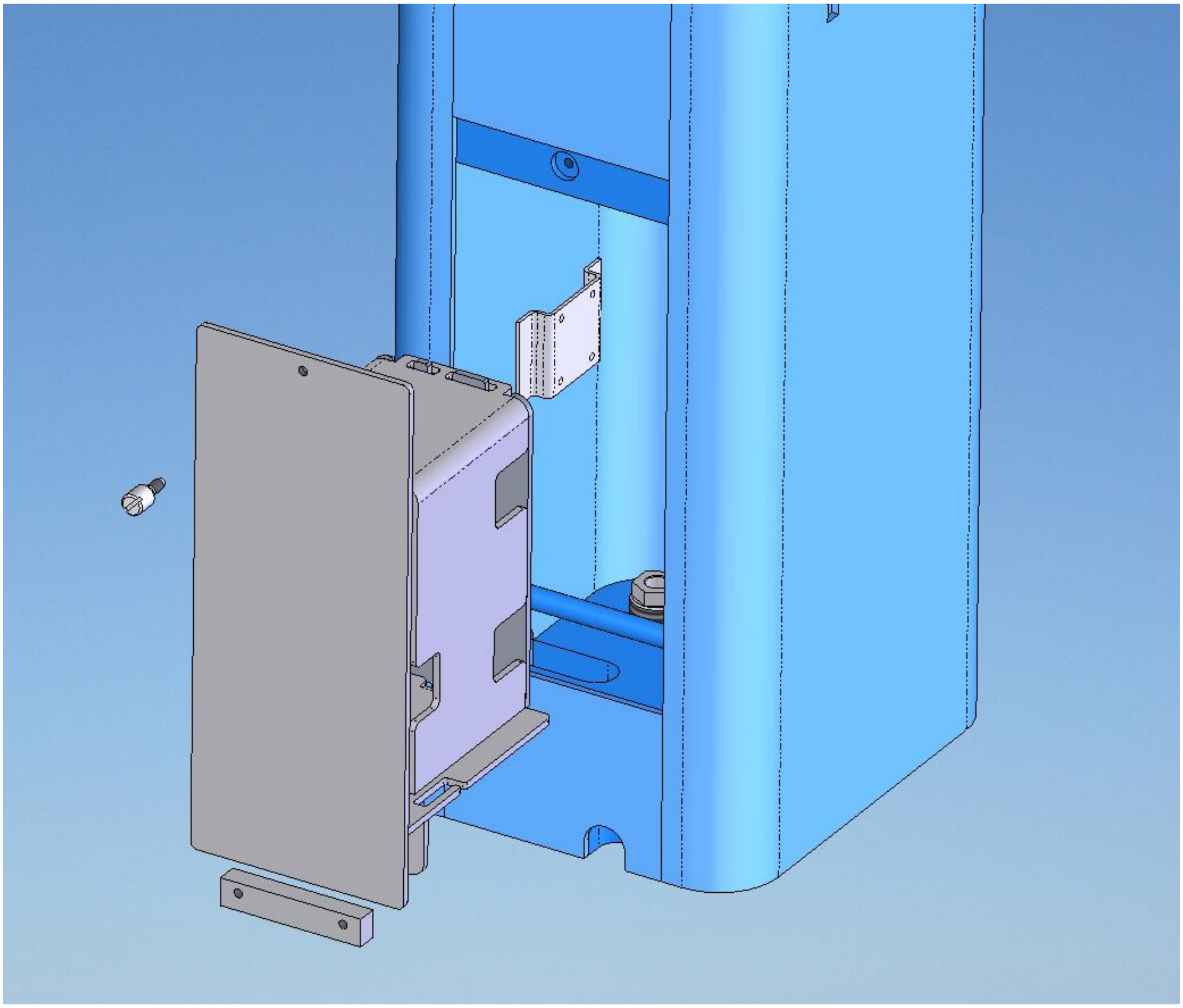


1. Note position of Spring Bumper Assembly from bumper face.
2. Unscrew Spring Bumper Assembly from Bumper.
3. Clean hole with alcohol swab and let dry.
4. Screw in new Spring Bumper Assembly to correct position.
5. Add a small amount of Loctite 290 and wipe off excess.

9.3 Access Door Assembly

Time Required: 50 minutes
Tools Required: 5/32” non-magnetic Allen wrench

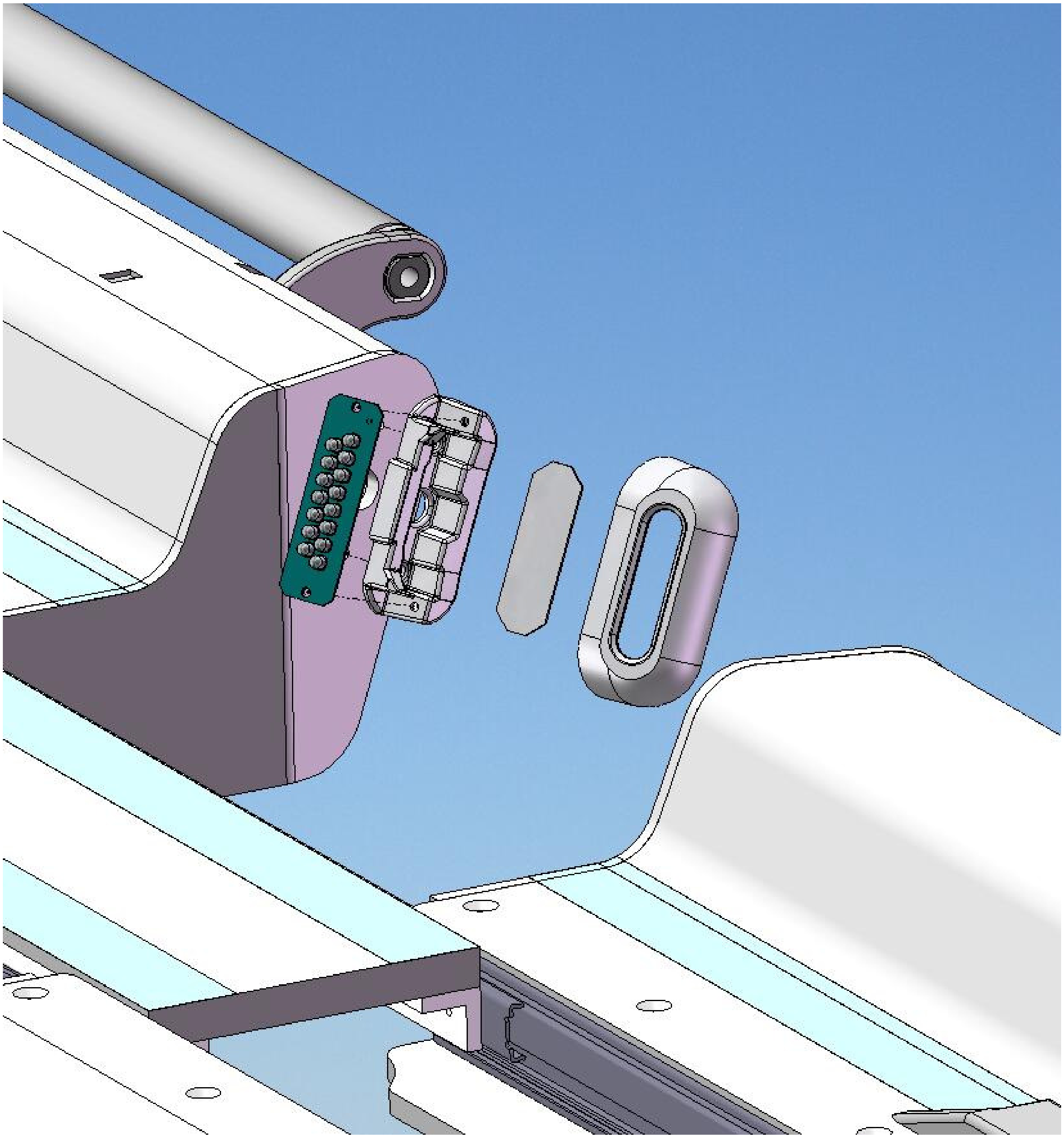
Parts Required: BEP-1000-FSA-005-000



1. Remove fuse from Battery PCB
2. Unplug Access Door Assembly from lighting system.
3. Unscrew block from bottom of enclosure assembly.
4. Insert new assembly on to mounting rod weldment and fasten block.
5. Plug into lighting system, insert fuse and test system.

9.4 Light Enclosure

Time Required: 40
Tools Required: Non-magnetic small slot screwdriver
Parts Required: GE0708-007



1. Remove Light Enclosure cover.
2. Replace Light Enclosure base.
3. Attach new Light Enclosure base and Light Enclosure cover.

9.5 Lighting Wiring Harness

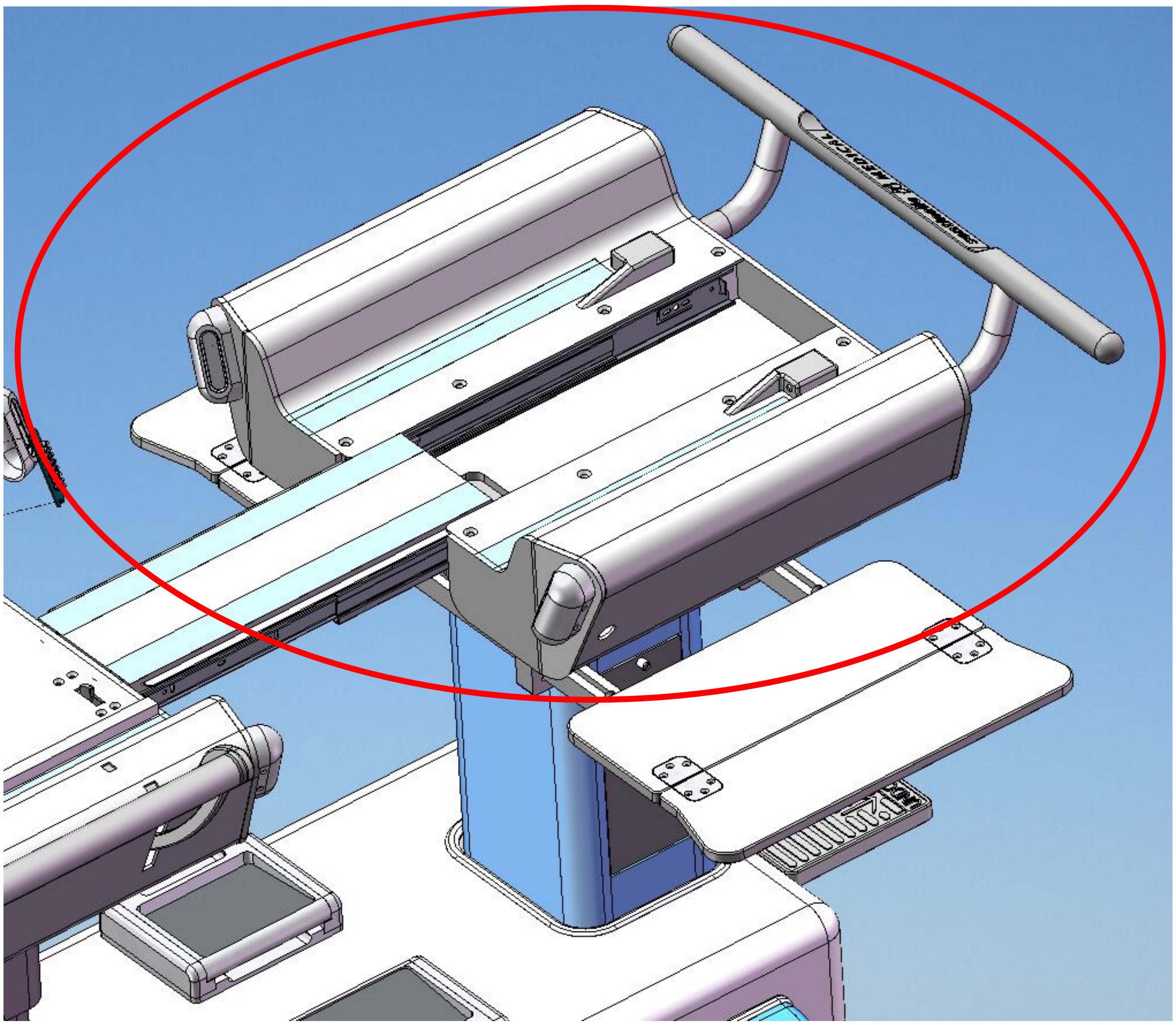
Time Required: 100 minutes
Tools Required: None
Parts Required: GE0708-001

1. Open Access Door Assembly to gain access to Lighting Wiring Harness.

2. Disconnect Lighting Wiring Harness from Light Enclosures in UBS and LBS.
3. Remove Lighting Wiring Harness from mounting tabs in FSA and RSA.
4. Remove Lighting Wiring Harness from conduit below Shroud.
5. Detach distribution PCB and remove Lighting Wiring Harness from FSA.
6. Attach distribution PCB and new Lighting Wiring Harness to FSA.
7. Route through conduit and attach new harness to tabs in FSA and RSA.
8. Connect Lighting Wiring Harness to Light Enclosures in UBS and LBS.

9.6 Upper Body Support

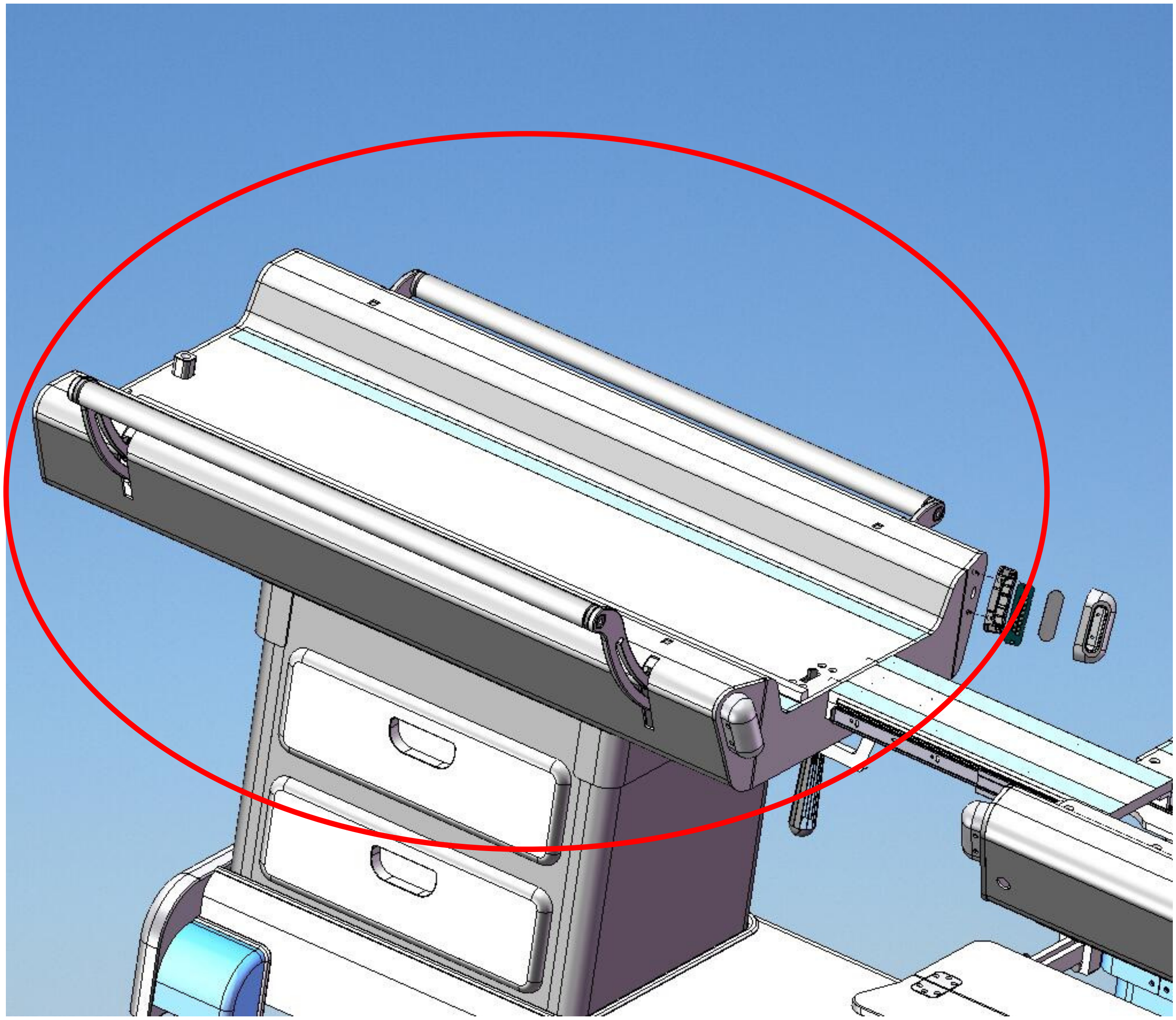
Time Required: 90
Tools Required: 9/16” non-magnetic wrench and 5/32” non-magnetic Allen key
Parts Required: BEP-1000-UBS-000-000



1. Detach Travel Stop Cable from No-Undock Hook.
2. Unplug Lighting Wiring Harness from Light Enclosures.
3. Remove four mounting bolts from inside of UBS.
4. Carefully lift off LBS.
5. Lower new LBS into position.
6. Pass Travel Stop Cable through FSA and let hang freely.
7. With bridge Guide Block aligned to LBS hole, fasten UBS with four mounting bolts from inside of UBS.
8. Attach Travel Stop Cable to No-Undock Hook and set hook height.
9. Plug Lighting Wiring Harness into Light Enclosures.

9.7 Lower Body Support

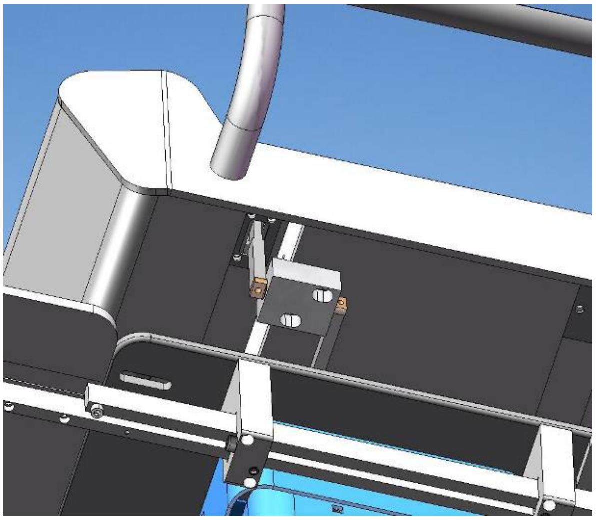
Time Required: 90 minutes
Tools Required: 9/16” non-magnetic wrench
Parts Required: BEP-1000-LBS-000-000

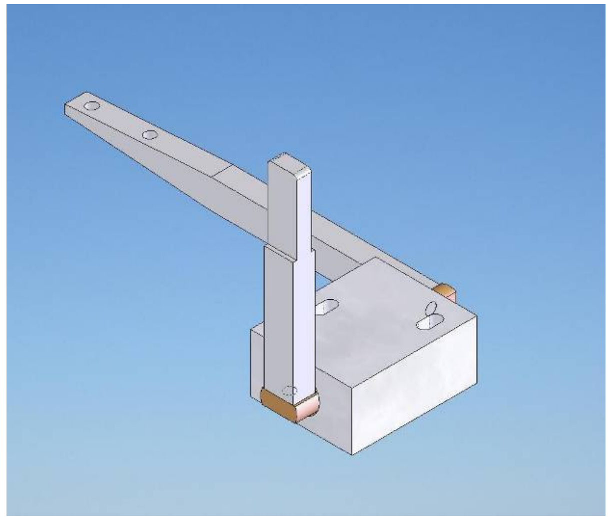


1. Remove drawers from RSA.
2. Detach Latch Tab Cable from Swing Assembly.
3. Unplug Lighting Wiring Harness from Light Enclosures.
4. Mark position of LBS on base and remove four mounting bolts from inside of LBS.
5. Carefully lift off LBS.
6. Lower new LBS into position.
7. Insert Latch Tab Cable through cable elbows.
8. Fasten LBS with four mounting bolts from inside of LBS in previously marked position.
9. Plug Lighting Wiring Harness into Light Enclosures.
10. Attach Latch Tab Cable and set tab height.
11. Replace drawers in RSA.

9.8 Travel Stop Pivot Assembly

Time Required: 50 minutes
Tools Required: 3/16” non-magnetic Allen wrench
Parts Required: BEP-1000-STP-000-000

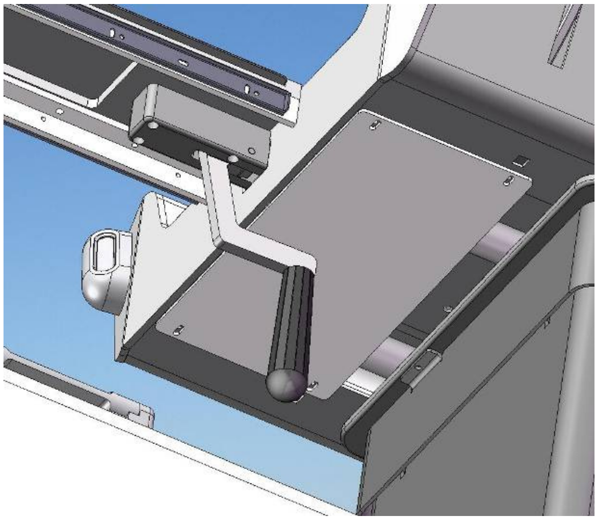
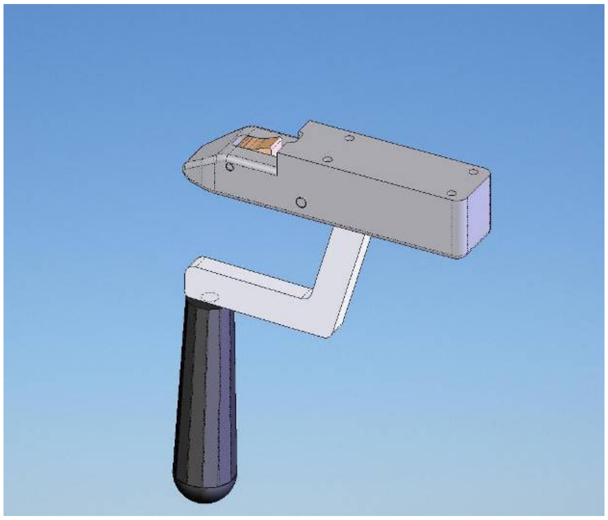




1. Detach Latch Travel Stop Cable from no undock hook.
2. Remove 2 mounting screws.
3. Carefully remove Travel Stop Pivot Assembly.
4. Thread new cable into pivot arm.
5. Mount Travel Stop Pivot Assembly and pass Stop Cable through FSA and let hang freely.
6. Fasten Travel Stop Pivot Assembly with 2 screws and align rocker arm with Travel Stop Plunger Assembly
7. Attach Travel Stop Cable to No-Undock Hook and set hook height.

9.9 Bridge Latch Assembly

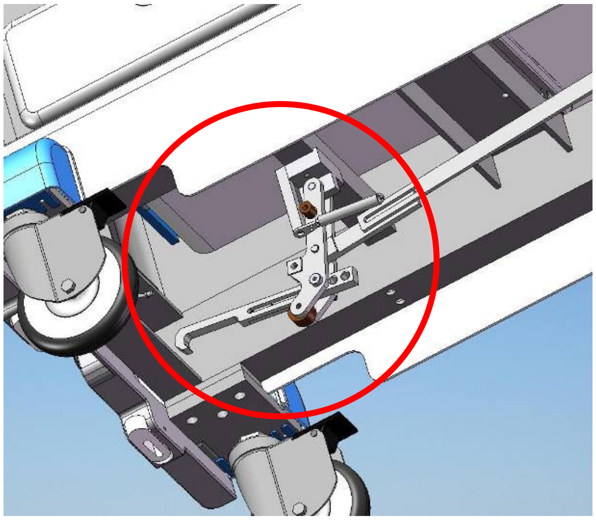
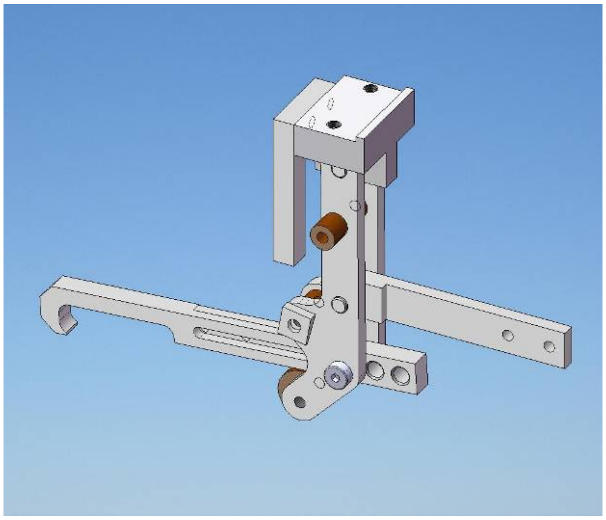
Time Required: 20 minutes
Tools Required: 3/16” non-magnetic Allen wrench
Parts Required: BEP-1000-BLA-000-000

1. Remove Bridge Latch Assembly by removing four mounting screws.
2. Replace Bridge Latch Assembly and fasten with four mounting screws.

9.10 Swing Assembly

Time Required: 70 minutes
Tools Required: ¼” non-magnetic Allen wrenches
Parts Required: BEP-1000-LNK-003-000

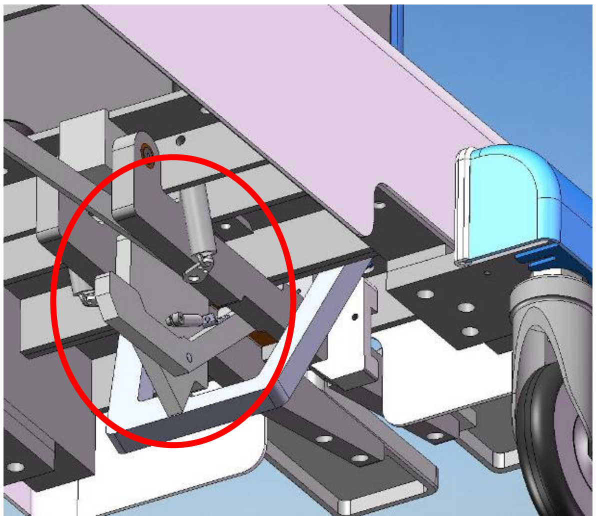



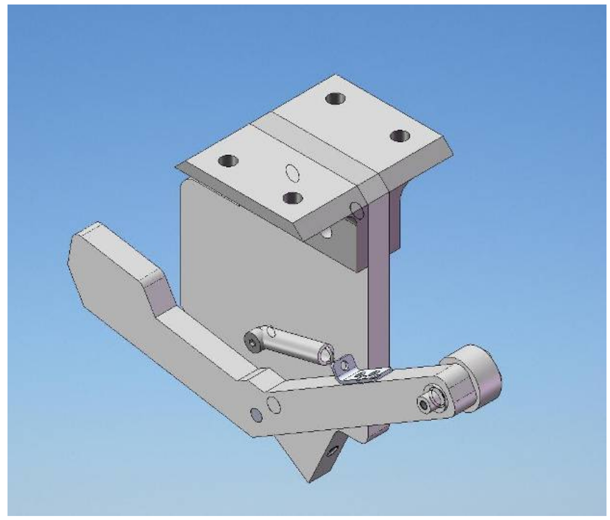
1. Remove drawers from RSA.
2. Detach Latch Tab Cable from Swing Assembly.
3. Detach Tension Coupling Docking Linkage from Swing Assembly by removing two bolts.
4. Detach Swing Assembly from Frame by removing mounting bolts.
5. Mount Swing Assembly to Frame with mounting bolts.
6. Attach Tension Coupling Docking Linkage to Swing Assembly by inserting two bolts while mounting

Undock Spring.
7. Attach Latch Tab Cable to Swing Assembly and refer to 7.15 to set Latch Tab Height.
8. Replace Drawers.

9.11 Docking Linkage Assembly

Time Required: 90 minutes
Tools Required: ¼” non-magnetic Allen wrench
Parts Required: BEP-1000-LNK-002-000

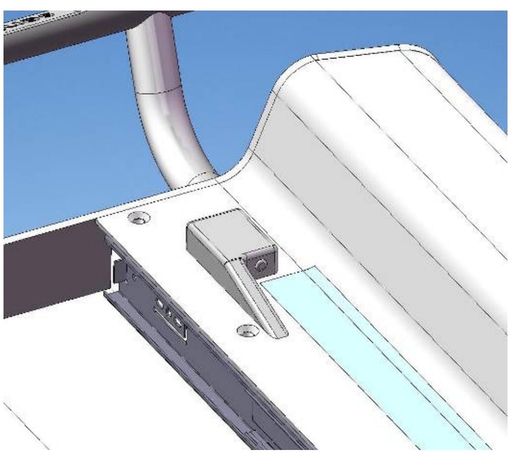


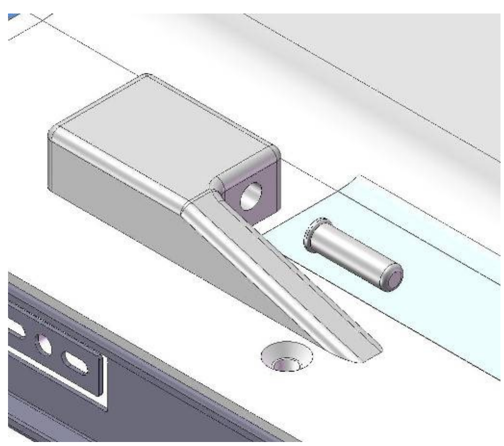


1. Detach Tension Coupling Docking Linkage from Link Wheel Pin by removing screw.
2. Remove Docking Linkage Assembly by removing four screws from mounting brackets.
3. Insert new Docking Linkage Assembly and fasten by inserting four screws into mounting brackets.
4. Attach Tension Coupling Docking Linkage to Link Wheel Pin with screw and Loctite 262.

9.12 Travel Stop Plunger Assembly

Time Required: 50 minutes
Tools Required: 3/16” non-magnetic Allen wrench
Parts Required: GE0708-002

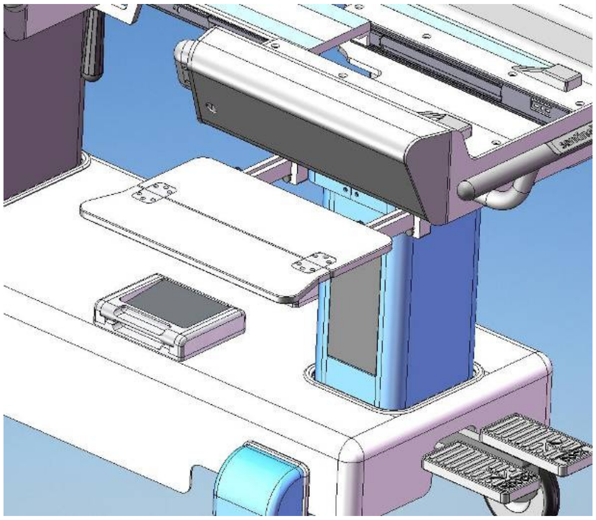




1. Remove Travel Stop Plunger Housing by removing four mounting screws from inside UBS.
2. Place new Travel Stop Plunger Assembly in position with plunger protruding on UBS and fasten with four mounting screws from inside UBS.

9.13 Tray Table

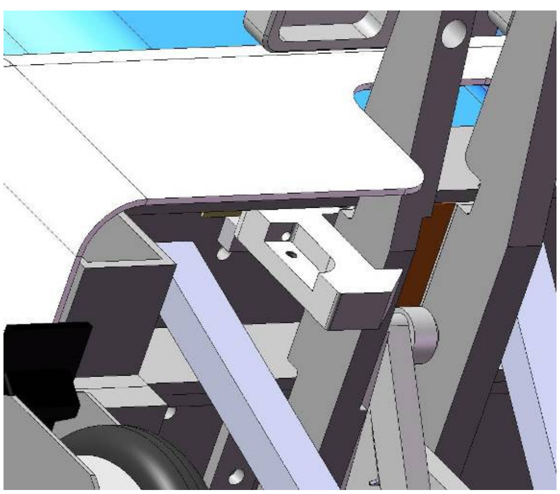
Time Required: 30 minutes
Tools Required: 5/32” non-magnetic Allen wrench
Parts Required: BEP-1000-FSA-004-000

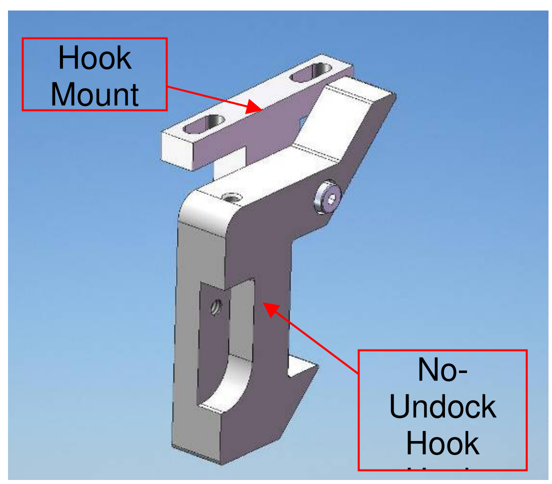


1. Remove Tray Assembly by removing 6 mounting screws from the underside.
2. Carefully position new Tray Assembly and fasten with 6 mounting screws from the underside using light force.

9.14 No-Undock Hook Assembly

Time Required: 70 minutes
Tools Required: 5/32” and 3/16” non-magnetic Allen wrench
Parts Required: GE0708-004

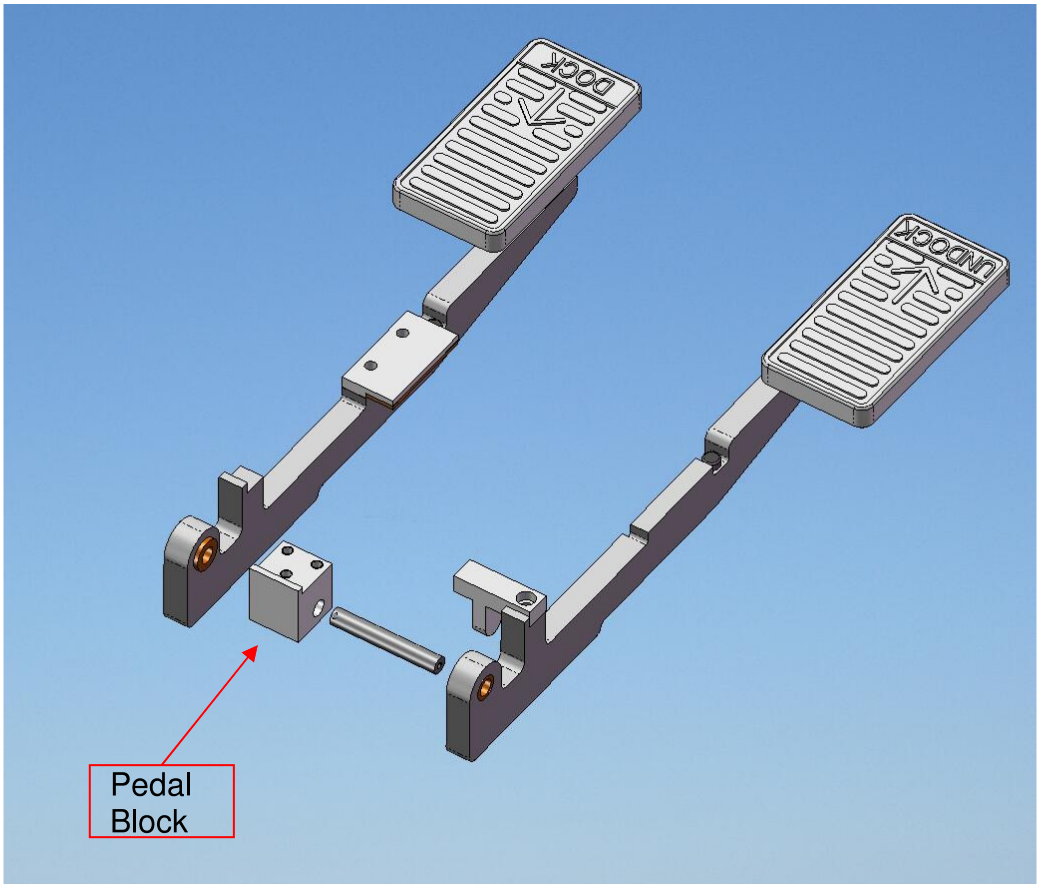




1. Remove two mounting screws from the No Undock Hook Mount and remove assembly.
2. Position new No-Undock Hook Assembly so the hook is positioned correctly with the pedal arm and fasten with two mounting screws in the No-Undock Hook Mount.

9.15 Pedal Block Assembly

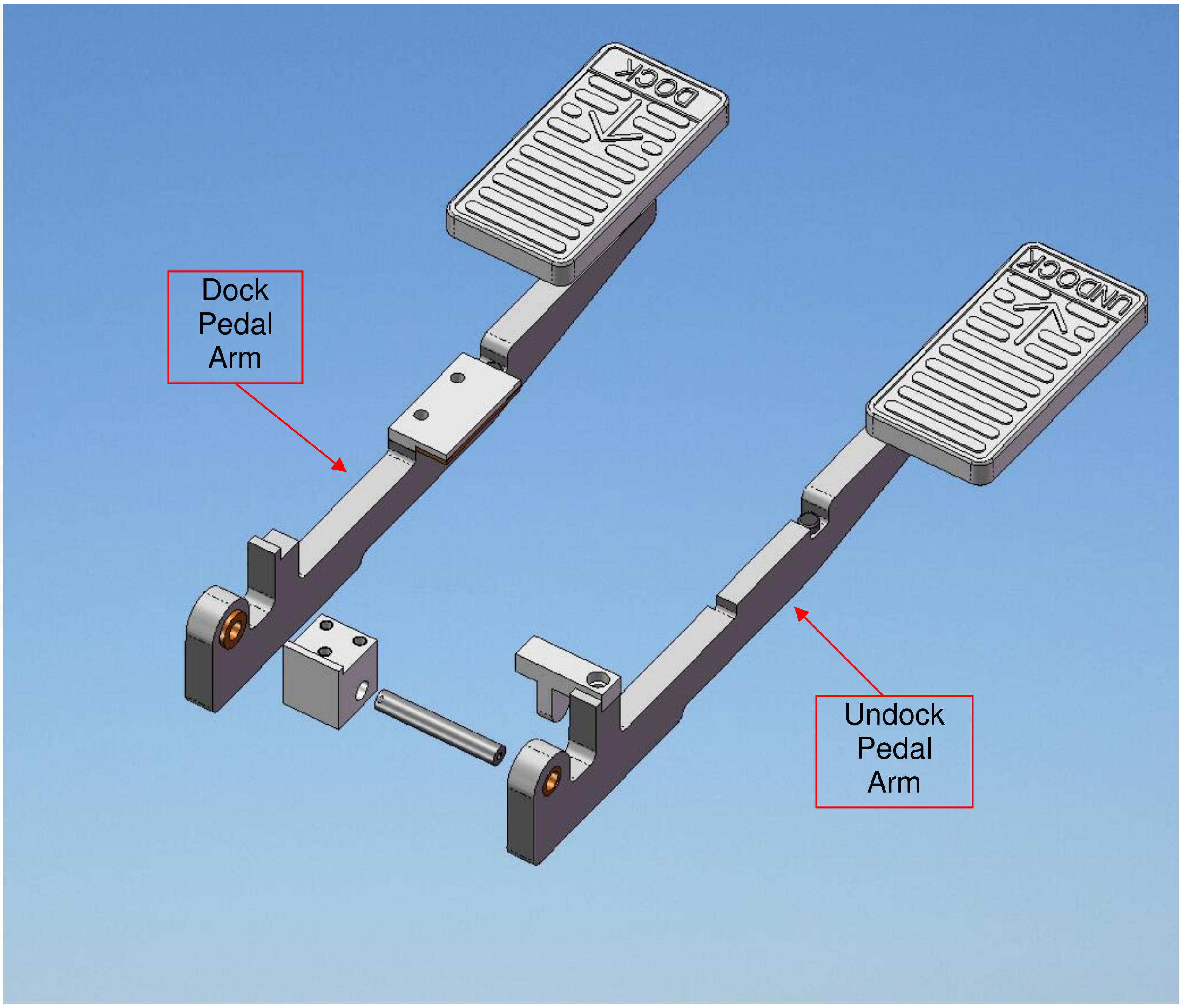
Time Required: 90 minutes
Tools Required: ¼”, 3/16”, 1/8” non-magnetic Allen wrench
Parts Required: GE0708-005



1. Remove Wheel Covers.
2. Remove Shroud mounting screws.
3. Raise shroud to gain access to Pedal Block Assembly and detach from frame by removing three mounting screws.
4. Remove Pedal Arms from Pedal Block Assembly by removing screws from each end of shaft.
5. Position new Pedal Block Assembly and fasten Pedal Arms to shaft with screws.
6. Fasten new Pedal Block Assembly to Frame with three mounting screws.
7. Lower Shroud into position and replace Shroud mounting screws.
8. Replace Wheel Covers.

9.16 Pedal Arm Assembly

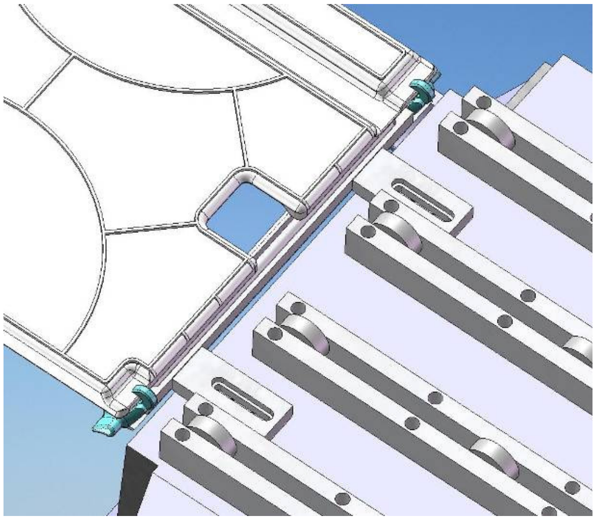
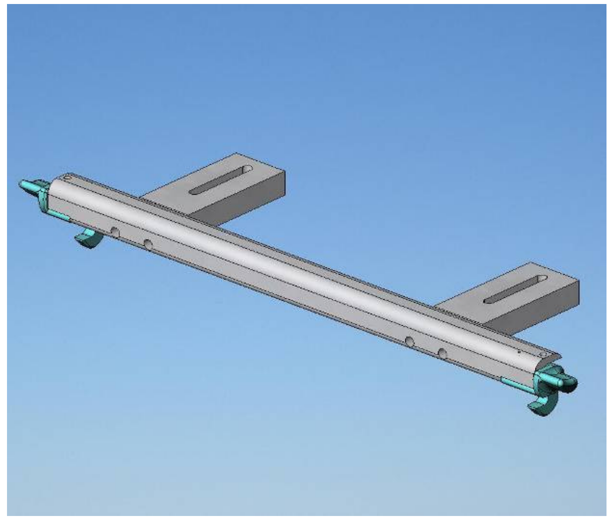
Time Required: 60 minutes
Tools Required: 1/8” non-magnetic Allen wrench
Parts Required: GE0708-006



1. Remove Pedal Arm from Pedal Block Assembly by removing screw from end of shaft.
2. Place new Pedal Arm on shaft and fasten with screw in end of shaft with Loctite 262.

9.17 Catchments Latch Assembly

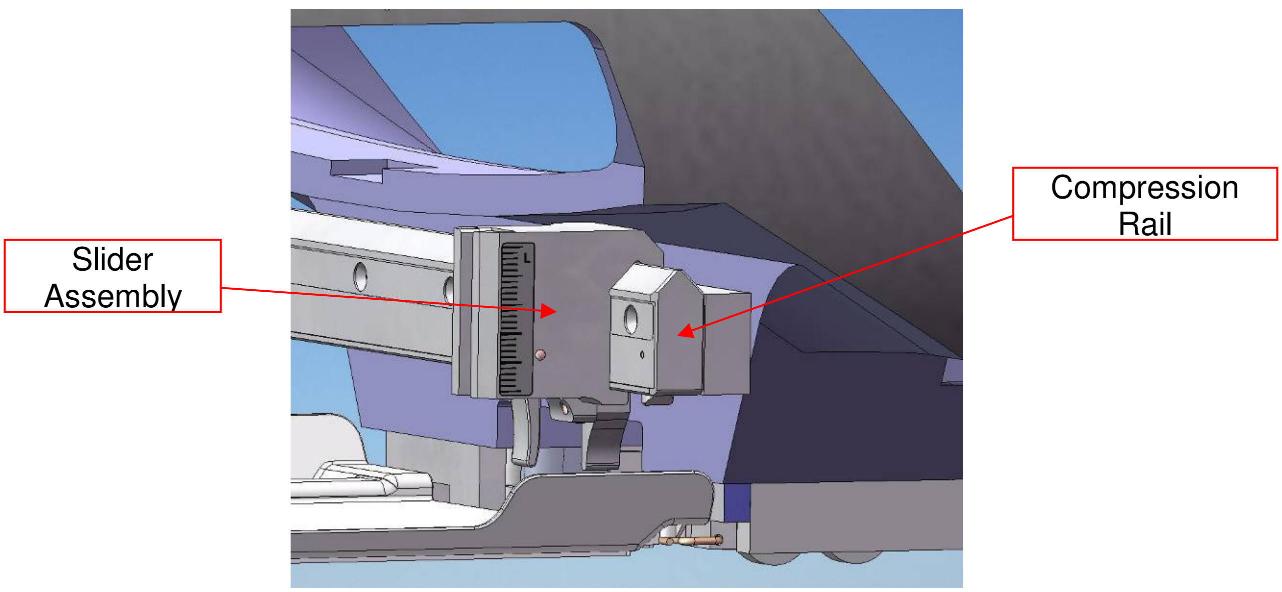
Time Required: 60 minutes
Tools Required: Medium non-magnetic slot screwdriver
Parts Required: 4000062-11

1. Access underside of Patient Support and loosen Door Catch Mount screws to remove Catchments Latch Assembly.
2. Replace Catchments Latch Assembly and tighten Door Catch Mount screws while aligning Catchments Latch Assembly to Catchments Tray.

9.18 Slider Assembly

Time Required: 40 minutes
Tools Required: none
Parts Required: 4000095-11

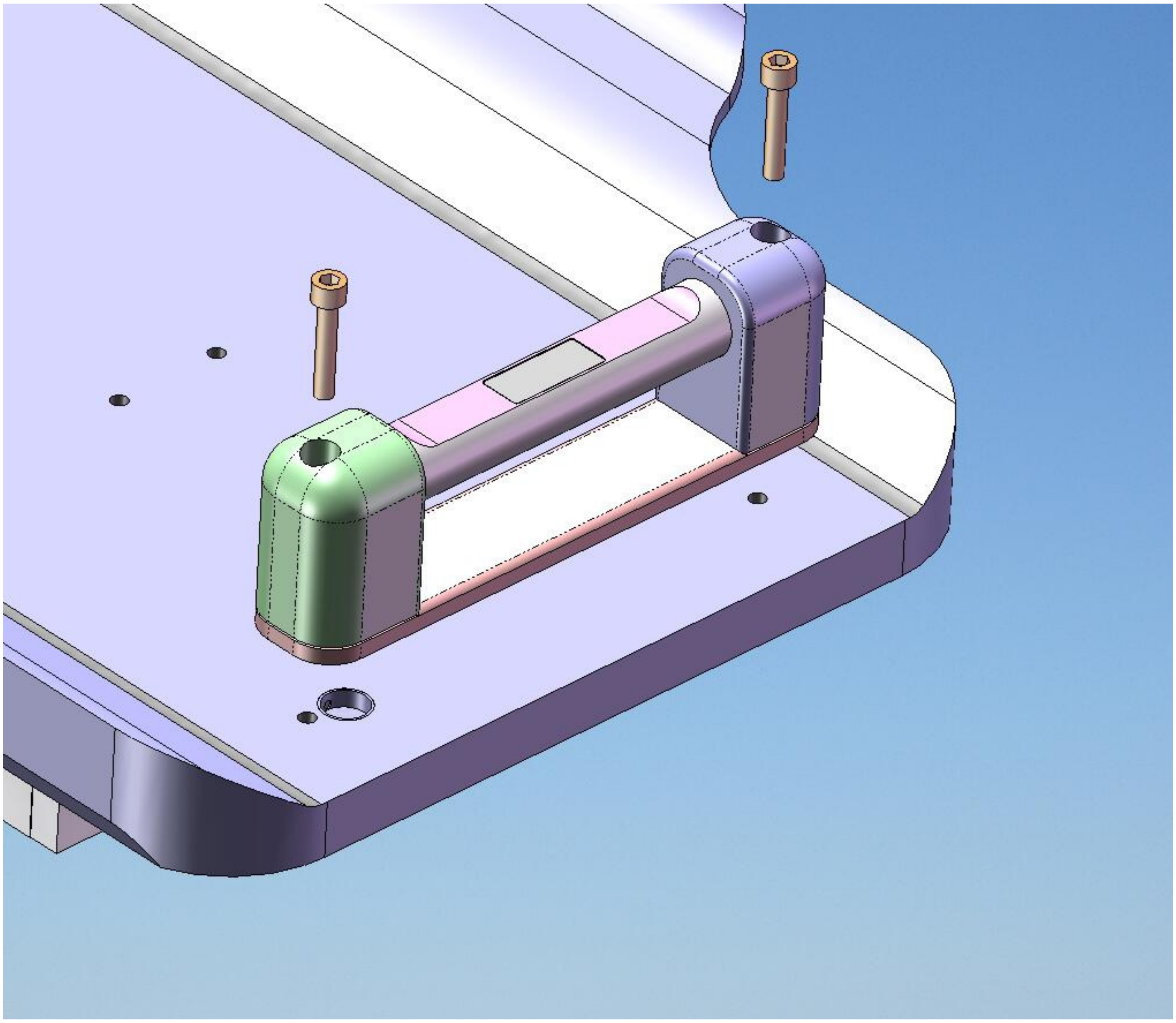


1. Depress retaining feature on end of Compression Rail and remove Slider.
2. Depress retaining feature on end of Compression Rail and replace Slider.
3. Slide Slider across entire length of rail to ensure functionality.
4. Test Slider with Compression Frame to ensure functionality.

9.19 Emergency Release Handle Assembly

Time Required: 90 minutes

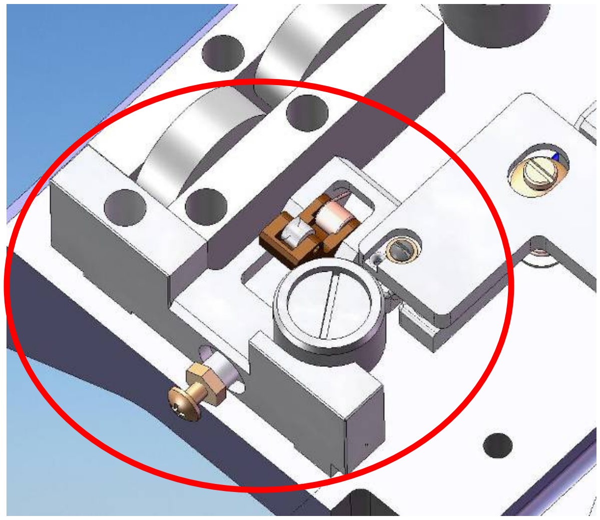
Tools Required: 3/16” non-magnetic Allen wrench
Parts Required: 4000031-11, Kevlar cord (10 feet)

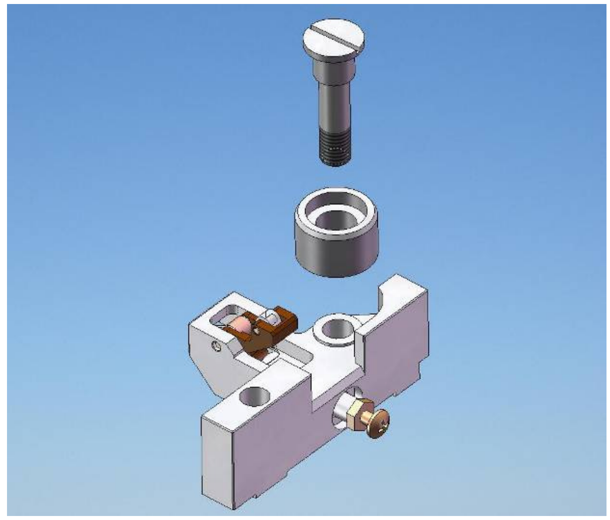


1. Remove Emergency Release Handle Assembly by removing two mounting screws.
2. Cut Kevlar cable as close to Emergency Release Handle Assembly as possible to leave excess.
3. Firmly knot Kevlar cable from inside patient support to Kevlar cable from new Emergency Release Handle Assembly and place Emergency Release Handle Assembly in position on patient support.
4. Access underside of Patient Support and pull Kevlar line through until taught.
5. Mount Emergency Release Handle Assembly on Patient Support with two screws.
6. Remove old Kevlar cable from Emergency Release Actuator and firmly knot new cable ensuring adequate tension and seal with tape.

9.20 Doghouse Mechanism Assembly

Time Required: 70 minutes
Tools Required: Non-magnetic adjustable wrench, non-magnetic large and medium slot screwdriver
Parts Required: 4000040-11

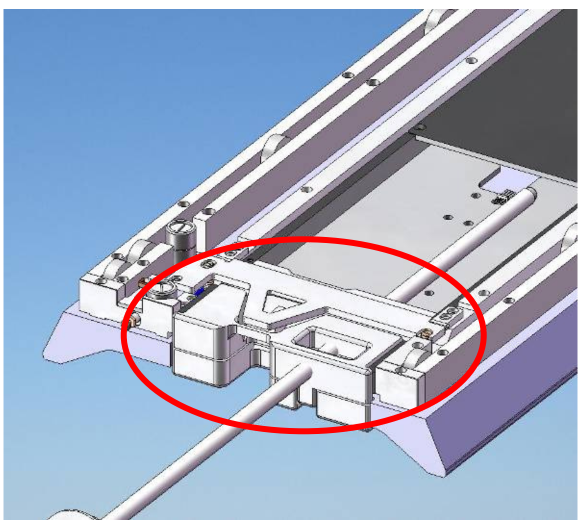


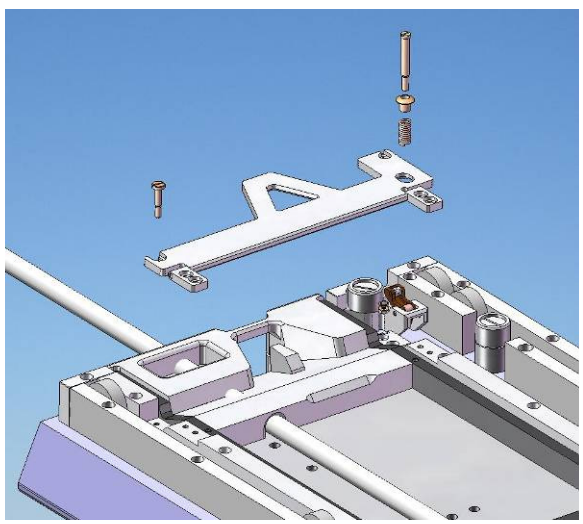


1. Measure protrusion distance of screw from plunger.
2. Access underside of Patient Support and remove L-R Centering Wheel Assembly and mounting screw and remove Doghouse Mechanism Assembly
3. Replace Doghouse Mechanism Assembly by fastening with L-R Centering Wheel Assembly and mounting screw.
4. Re-set Doghouse Mechanism by adjusting screw to previously measured setting and tighten locking nut firmly.

9.21 Emergency Release Spring Assembly

Time Required: 70 minutes
Tools Required: Non-magnetic small slot screwdriver
Parts Required: 4000104-11

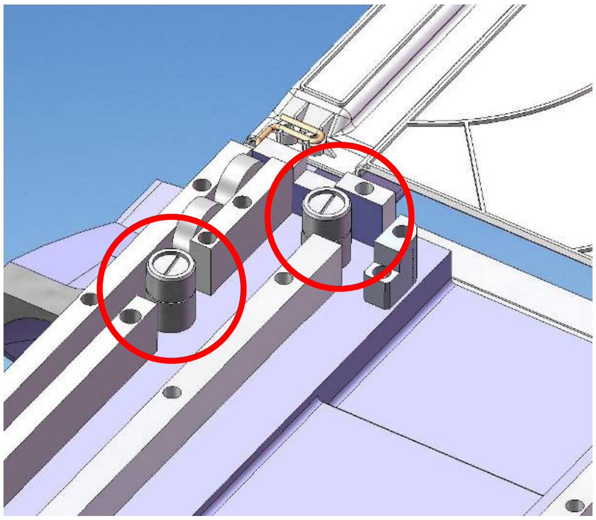


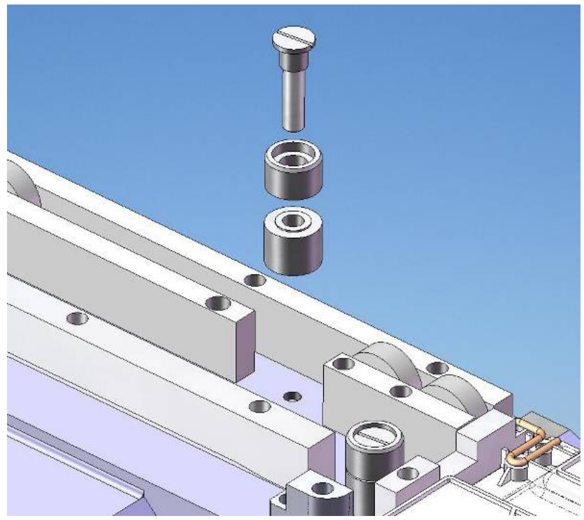


1. Access underside of Patient Support and remove mounting screws of Emergency Release Spring Assembly.
2. Remove tape and untie knotted Kevlar cable from Emergency Release Spring Assembly.
3. Position new Emergency Release Spring Assembly and fasten to Patient Support with mounting screws.
4. Firmly knot Kevlar cable to Emergency Release Actuator ensuring adequate tension and seal with tape.

9.22 L-R Centering Wheel Assembly

Time Required: 40 minutes
Tools Required: Non-magnetic large slot screwdriver
Parts Required: 4000017-11

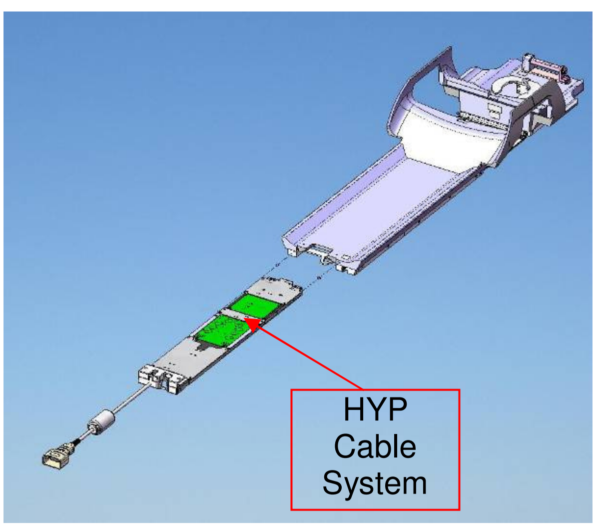


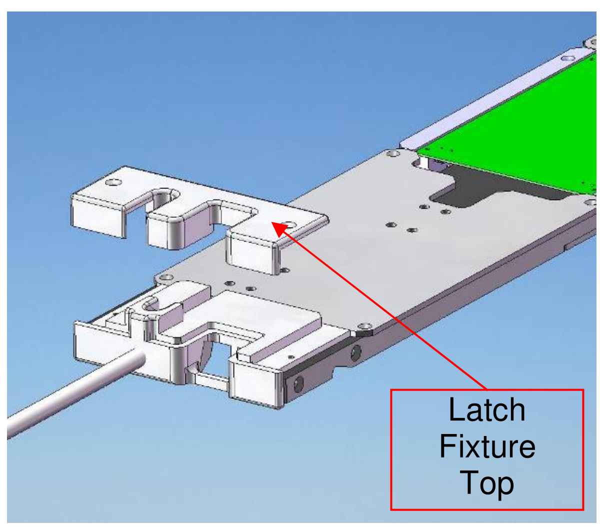


1. Access underside of Patient Support and unscrew mounting screw to remove L-R Centering Wheel Assembly.
2. Position replacement assembly and fasten carefully to underside of Patient Support.

9.23 Vanguard HYP Cable System

Time Required: 60 minutes
Tools Required: Non-magnetic large slot screwdriver
Parts Required: 4000015-11

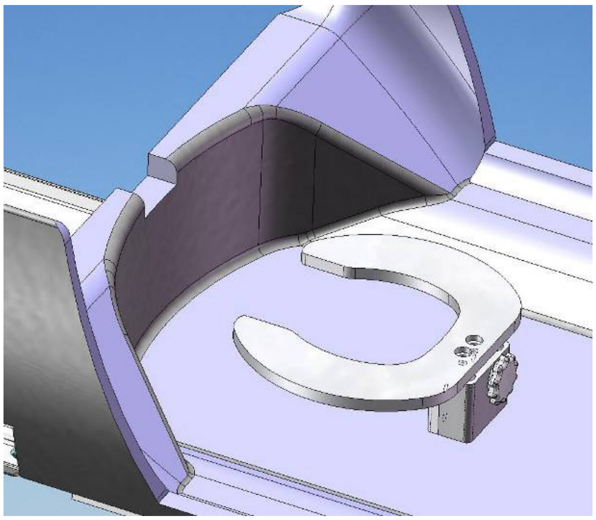


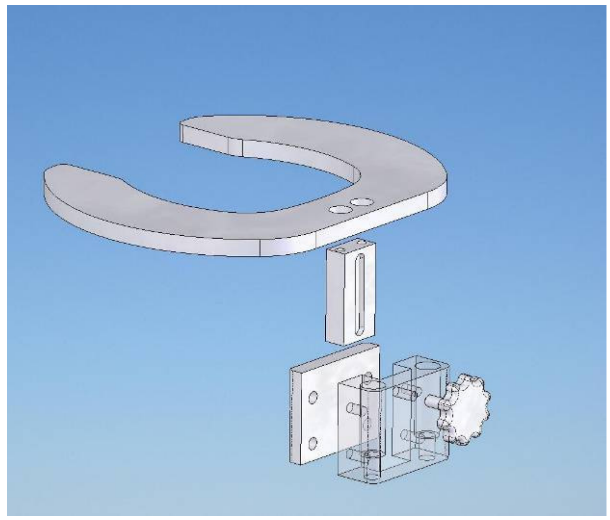


1. Remove latch fixture top by unscrewing two screws.
2. Remove Vanguard HYP Cable System from Patient Support by pulling on the bottom of the latch fixture.
3. Insert new Cable Tray into end of Patient Support and carefully slide completely in, positioning connectors at other end as required.
4. Position latch fixture top and fasten with two screws.

9.24 Head Rest Assembly

Time Required: 10 minutes
Tools Required: ¼" non-magnetic Allen wrench
Parts Required: 4000002-11

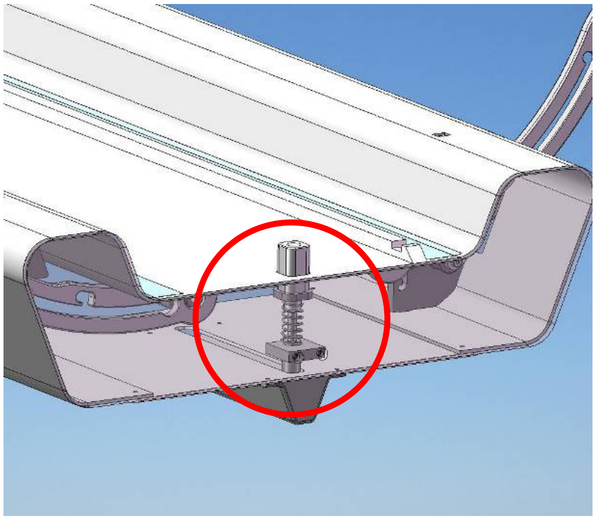


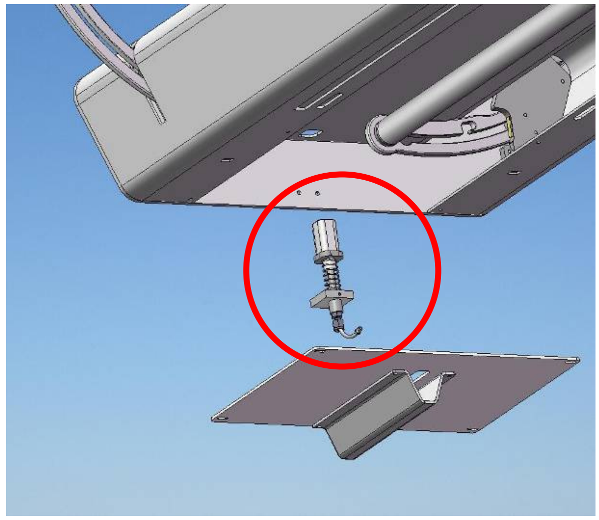


1. Remove Head Rest Assembly by unscrewing two mounting screws.
2. Position new Head Rest Assembly on Patient Support, insert mounting screws into new Head Rest Assembly and fasten firmly.

9.25 Latch Tab Assembly

Time Required: 40 minutes
Tools Required: Non-magnetic large slot screwdriver, 3/16" and 1/8" Allen wrench
Parts Required: BEP-1000-TAB -000-000

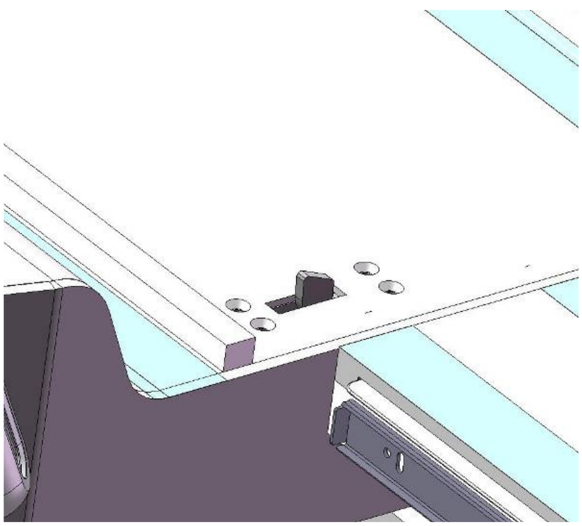


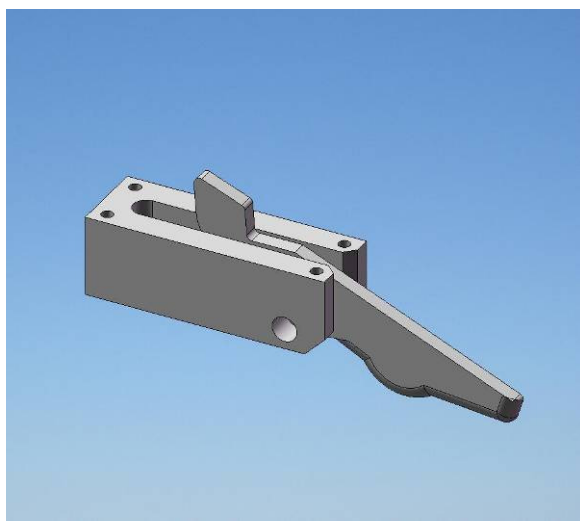


1. Remove cover by unscrewing four mounting screws while supporting cover.
2. Detach Latch Tab Cable from Swing Assembly.
3. Remove screw from top of Latch Tab and withdraw cable.
4. Insert new cable into new Latch Tab Assembly and through cable elbows.
5. Mount Latch Tab Assembly to LBS by screwing in two fasteners.
6. Position cover by screwing in four mounting screw while supporting cover.
7. Refer to 7.15 to set Latch Tab.

9.26 Stopper Assembly

Time Required: 40 minutes
Tools Required: Non-magnetic 5-32” Allen wrench
Parts Required: GE0708-003



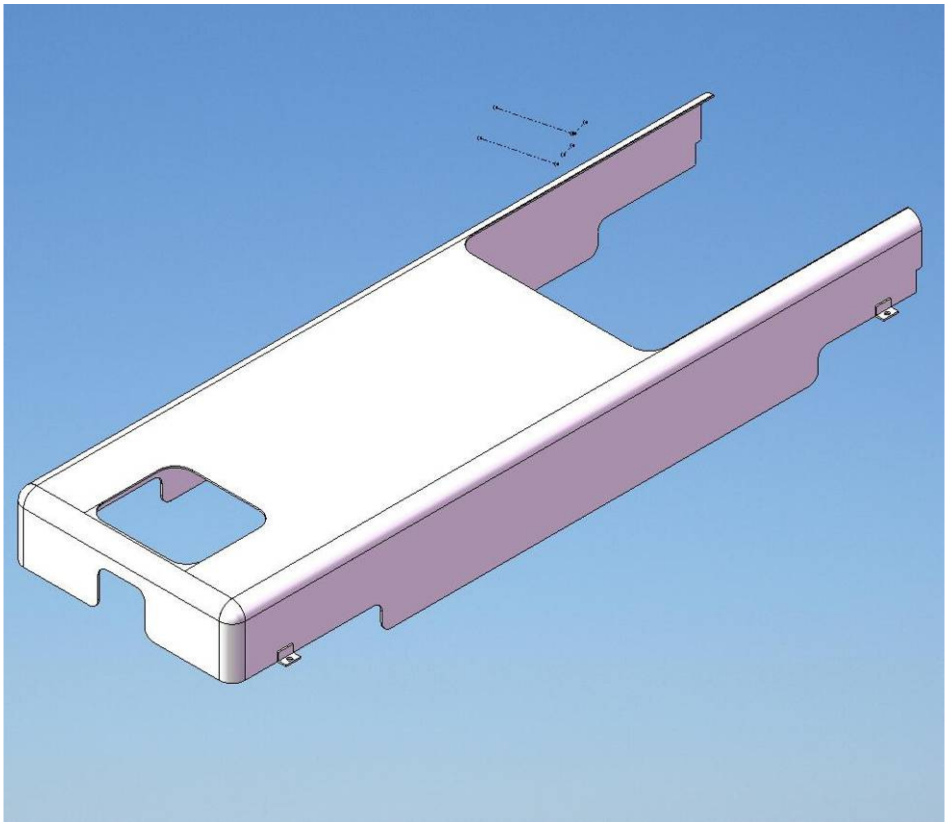


10.0 Replacing Parts

10.1 Shroud

Time Required: 120 minutes

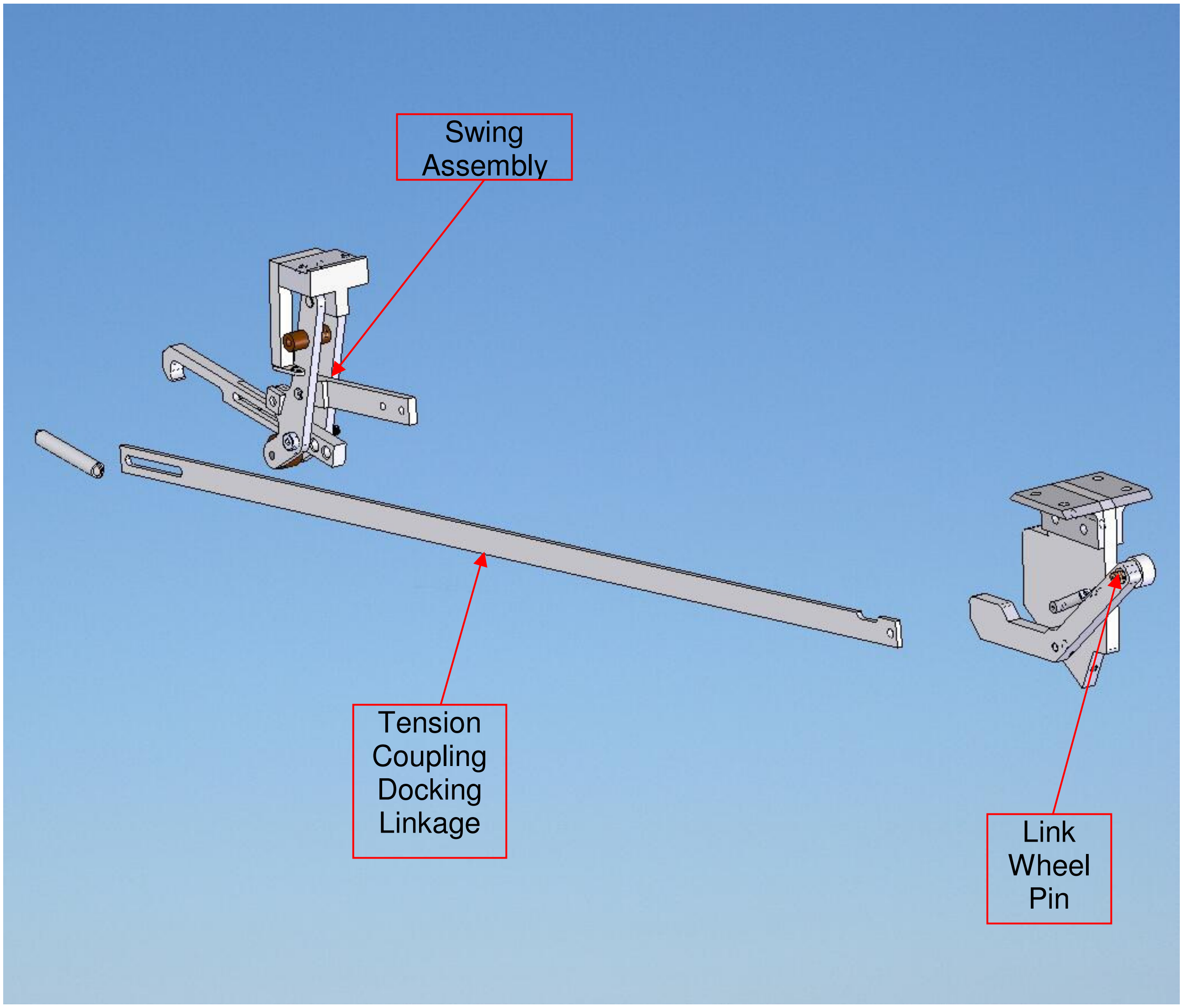
Tools Required: 9/16” non-magnetic wrench, 5/13” and 3/16” non-magnetic Allen key.
Parts Required: BEP-1000-FRM-001-000



1. Loosen mounting screws and remove all 4 Wheel Covers.
2. Remove Shroud mounting screws.
3. Remove UBS (refer to section 9.6)
4. Lift Shroud off of Frame.
5. Attach UBS (refer to section 9.6)
6. Fasten Shroud mounting screws.
7. Replace Wheel Covers and tighten mounting screws.

10.2 Tension Coupling Docking Linkage

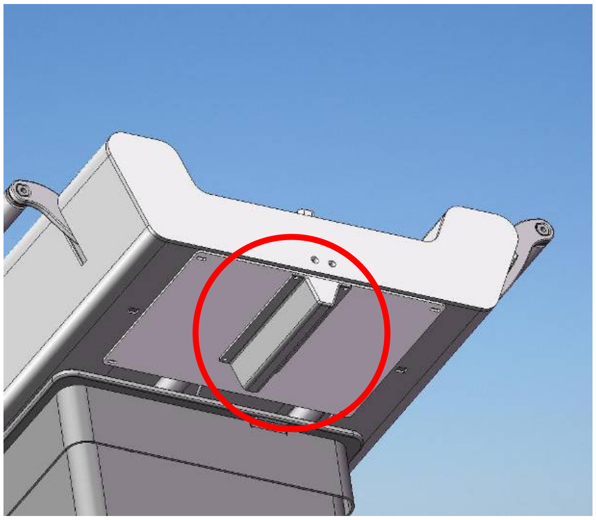
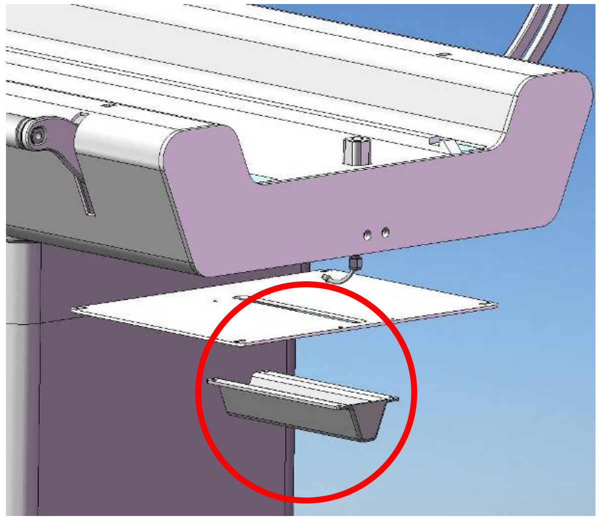
Time Required: 50 minutes
Tools Required: ¼”, 5/32” and 1/8” non-magnetic Allen wrench
Parts Required: BEP-1000-LNK-000-001



1. Unscrew mounting screw and remove Tension Coupling Docking Linkage from Link Wheel Pin.
2. Remove Tension Coupling Docking Linkage from Swing Assembly by removing the two mounting/adjustment screws.
3. Attach the new Tension Coupling Docking Linkage to the Swing Assembly by screwing in the mounting/adjustment screws.
4. Attach the Tension Coupling Docking Linkage to the Link Wheel Pin with the mounting screw and Loctite 262.

10.3 Cable Elbow Guard

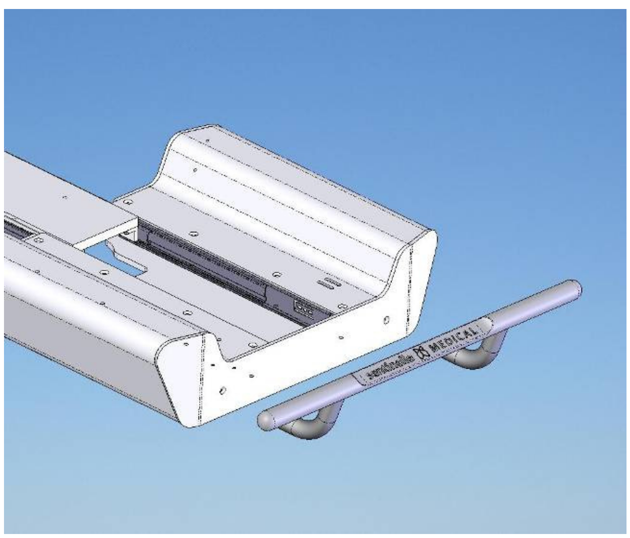
Time Required: 15 minutes
Tools Required: 1/8” non-magnetic Allen wrench
Parts Required: 5000242-11

1. Unscrew four mounting screws to remove Cable Elbow Guard.
2. Position new Cable Elbow Guard and fasten with four mounting screws.

10.4 UBS Handle

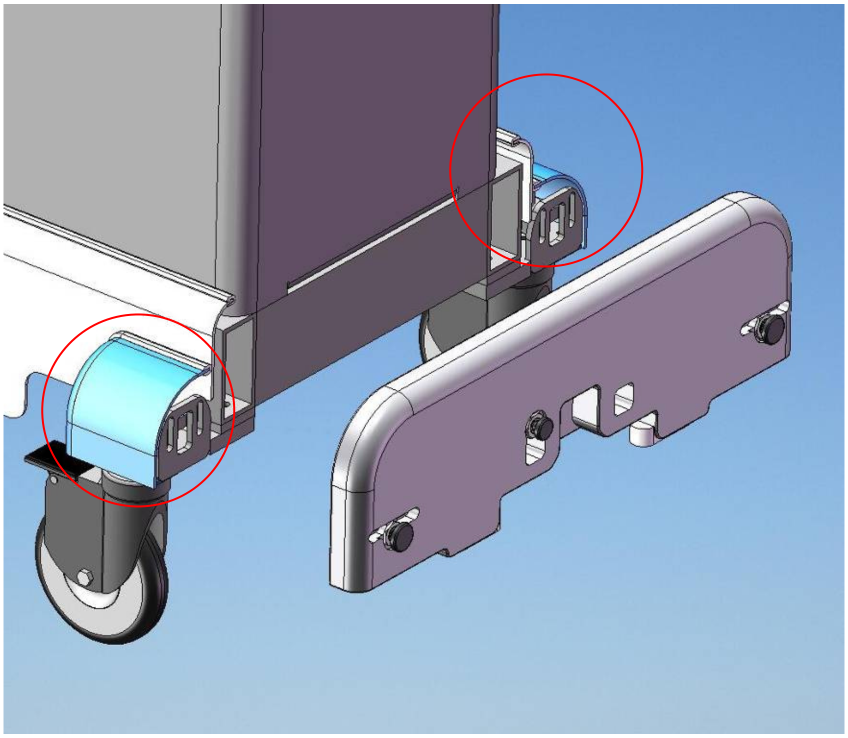
Time Required: 20 minutes
Tools Required: ½” non-magnetic adjustable wrench
Parts Required: BEP-HAN-000-001



1. Support UBS Handle and remove two mounting screws from inside of UBS.
2. Position and support new UBS Handle and firmly tighten mounting screws from inside of patient support.

10.5 Bumper

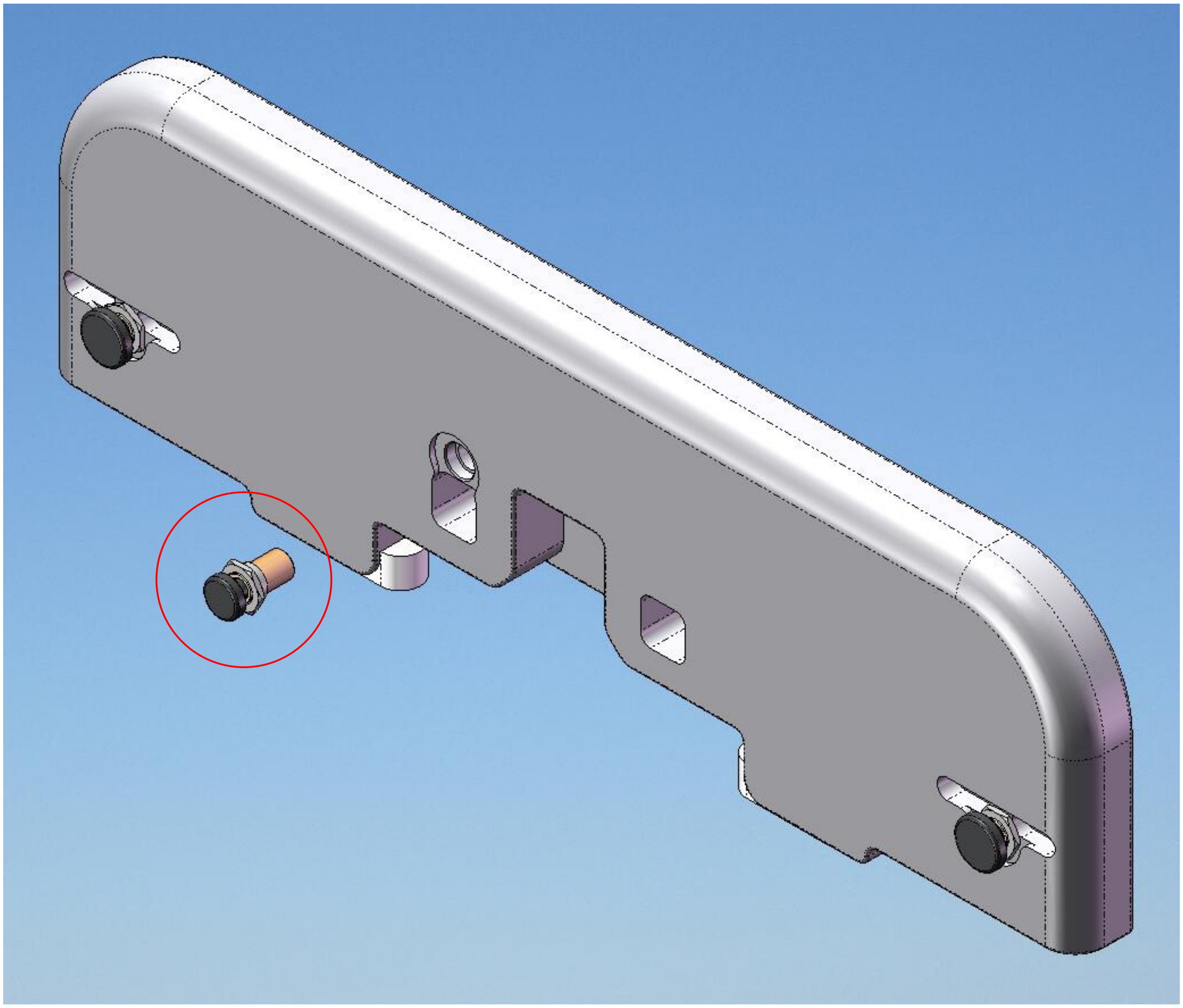
Time Required: 35 minutes
Tools Required: ¼” non-magnetic Allen key
Parts Required: BEP-1000-FRM-005-001



1. Loosen mounting screws and remove the 2 Wheel Covers next to the Bumper.
2. While supporting the bumper carefully unscrew the four mounting screws and remove the bumper.
3. Take measure the amount of protrusion of the Bumper Spring Assemblies and Spacer pin bumpers and screw them into the new bumper the same amount and apply a small amount of Loctite 290.
4. Attach new bumper to Frame by inserting mounting screws into bumper and screwing into tightening blocks.

10.6 Spacer Pin Bumper

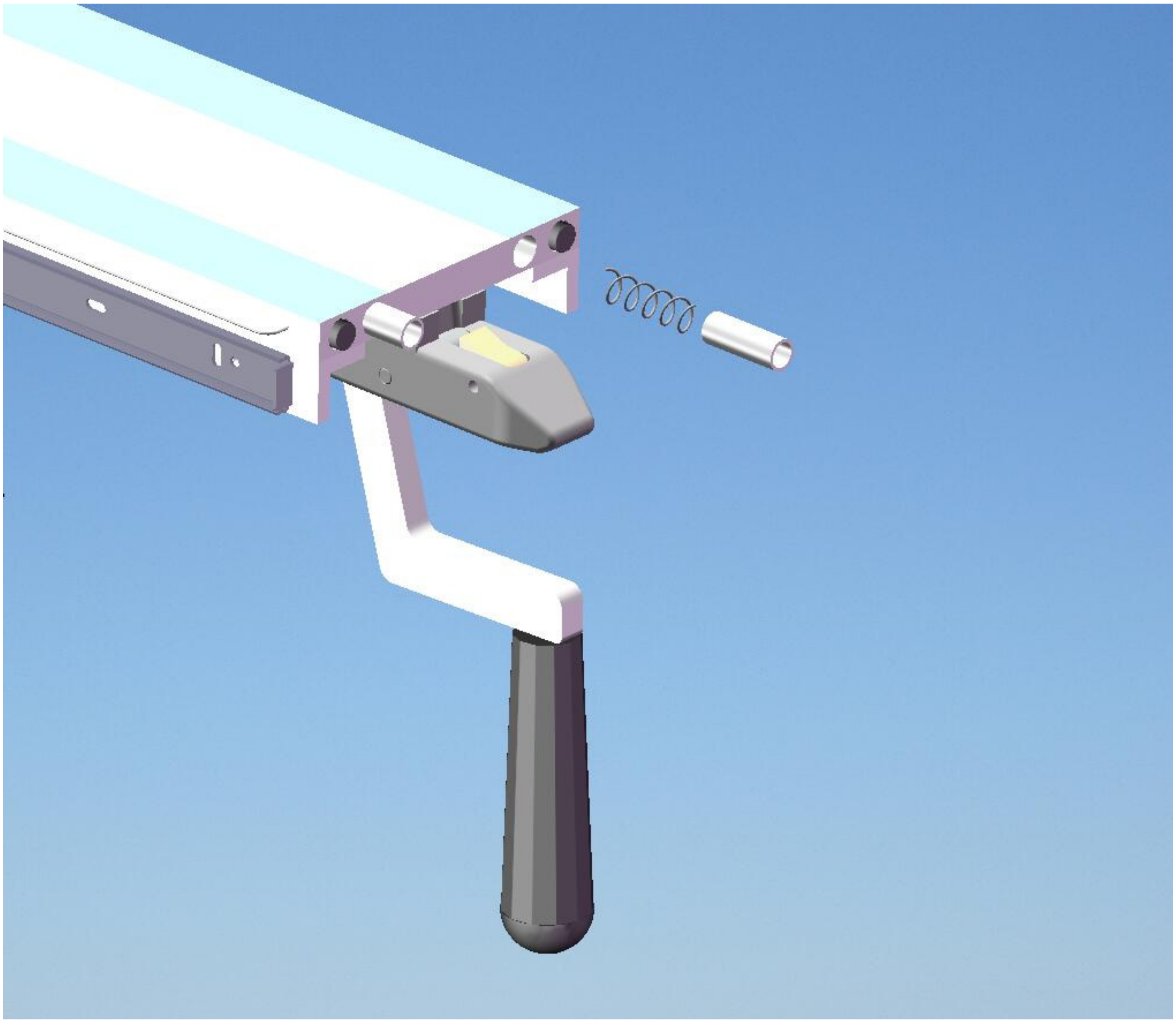
Time Required: 15 minutes
Tools Required: Non-magnetic adjustable wrench
Parts Required: BEP-10000-FRM-005-002



1. Measure protrusion distance of Spacer pin bumper from Bumper and remove.
2. Screw in new Spacer pin bumper to the correct amount to preserve the setting and apply a small amount of Loctite 290.

10.7 Bridge Spring Block

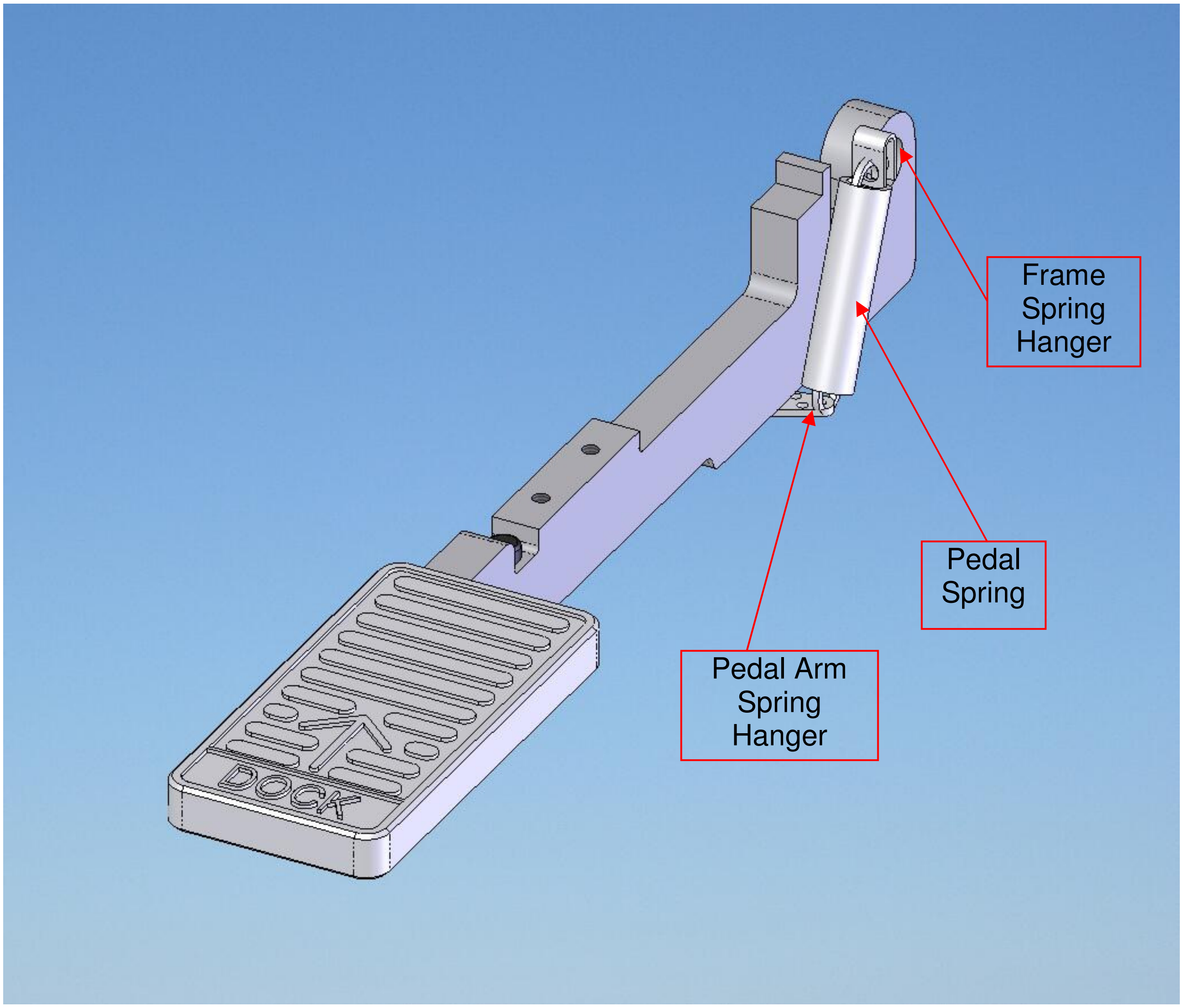
Time Required: 40 minutes
Tools Required: 3/16" non-magnetic Allen wrench
Parts Required: 4000104-11



1. With bridge in open position unscrew countersunk screw inside Spring Block and remove screw, spring and Spring Block.
2. Insert new spring, spring block and screw and tighten screw.

10.8 Pedal Spring

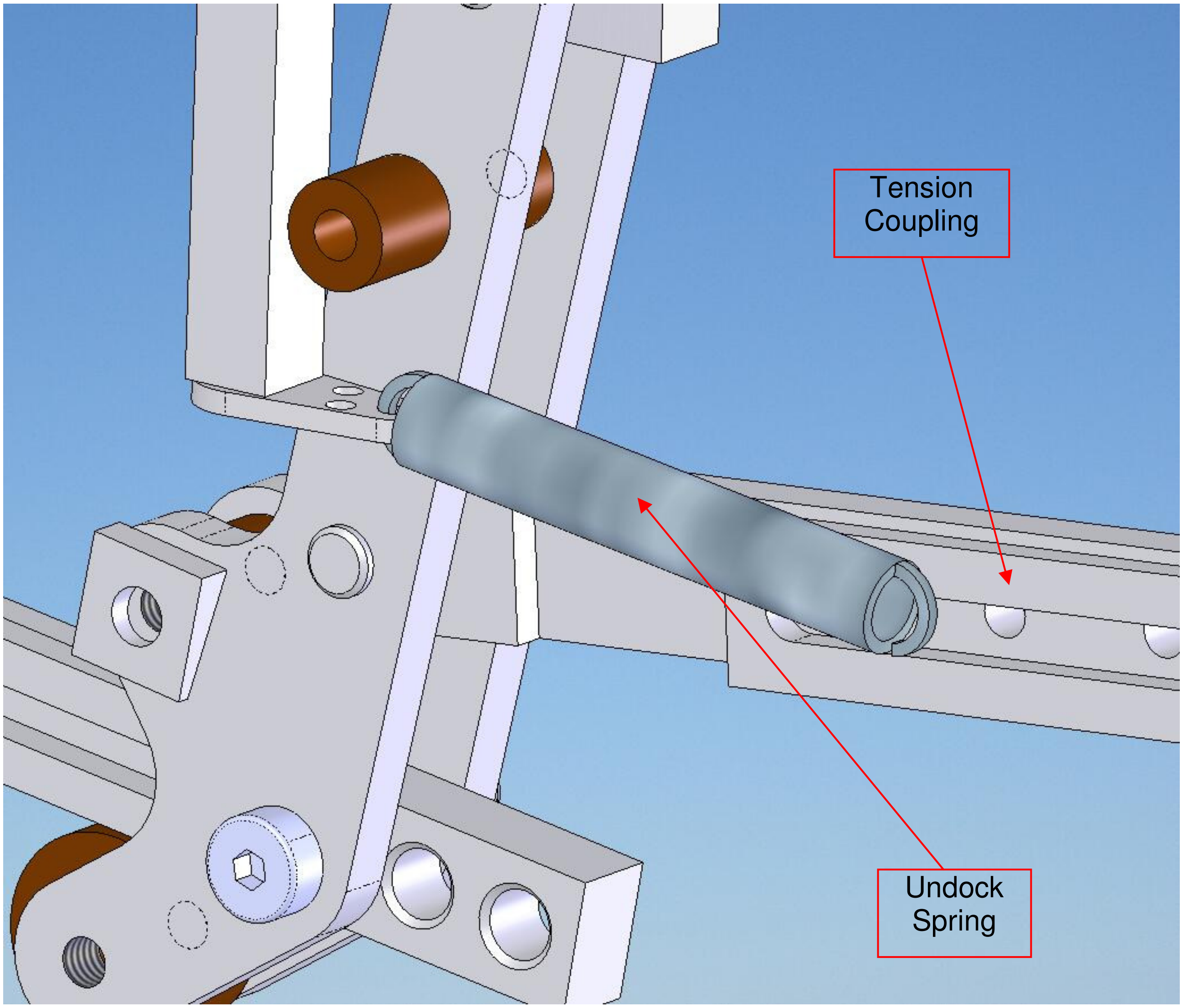
Time Required: 80 minutes
Tools Required: 1/4” non-magnetic Allen wrench
Parts Required: 4000104-11



1. Remove spring hanger by unscrewing screw on bottom of pedal arm.
2. Remove spring hanger on frame by unscrewing screw on frame.
3. Place new spring on hanger and fasten to frame with mounting screw.
4. Attach spring hanger to new spring and stretch spring to position hanger on Pedal Arm.
5. Fasten to Pedal Arm by screwing in mounting screw.

10.9 Undock Spring

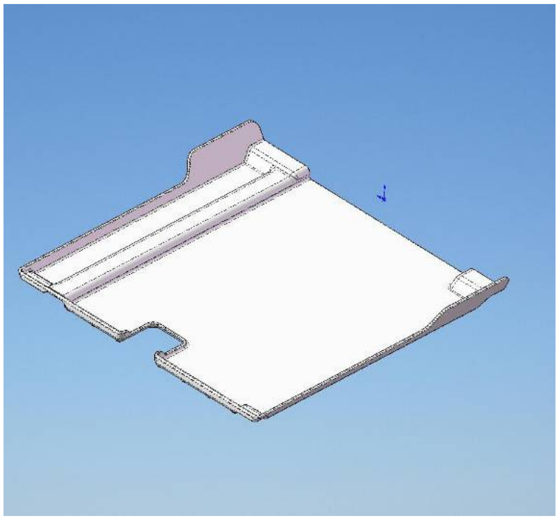
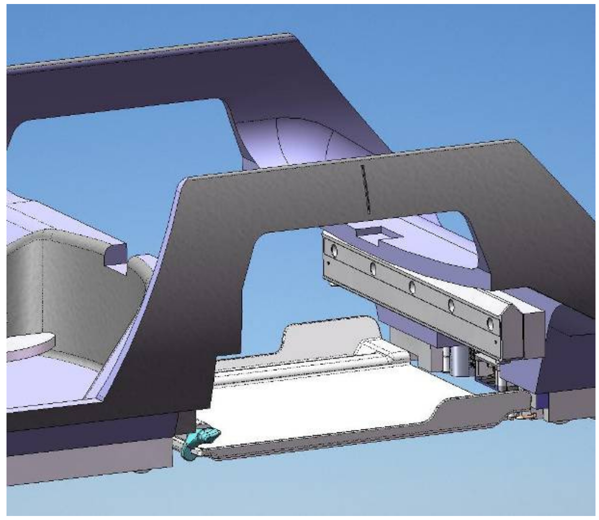
Time Required: 40 minutes
Tools Required: ¼” and 5/32” non-magnetic Allen wrench
Parts Required: 4000104-11



1. Remove Undock Spring Hanger from Swing Assembly by unscrewing mounting screw.
2. Remove Undock Spring from Tension Coupling Docking Linkage by unscrewing mounting screw and removing nut.
3. Attach new Undock Spring to Spring Hanger and attach to Swing Assembly by firmly screwing in mounting screw.
4. Insert mounting screw into other end of Undock Spring, screw nut on to halfway point of mounting screw and screw into Tension Coupling Docking Linkage so that end of screw is flush with far side of Tension Coupling Docking Linkage.

10.11 Catchment

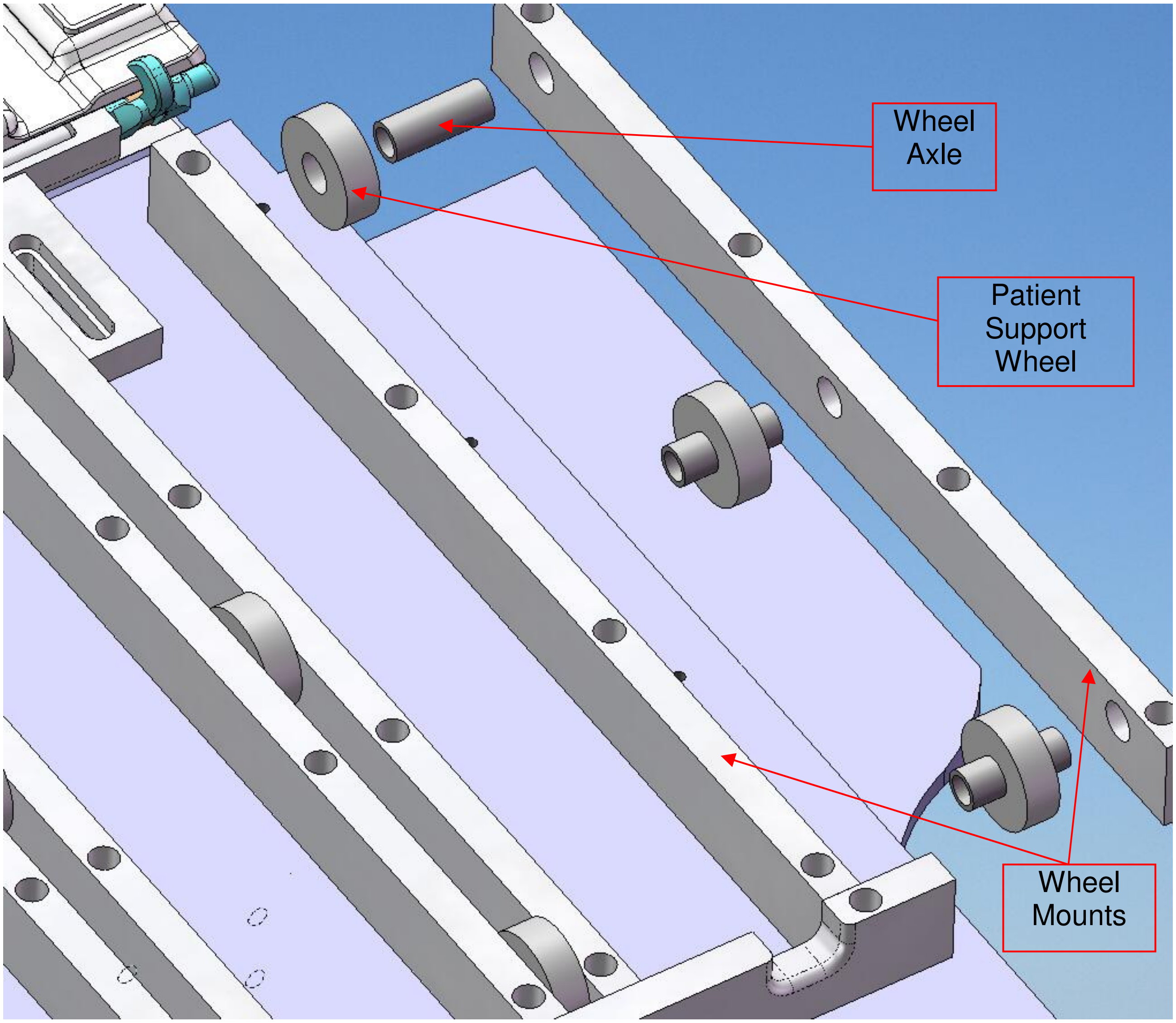
Time Required: 40 minutes
Tools Required: Non-magnetic small and medium slot screwdriver
Parts Required: 5000182-11, Southco mini offset hinge A 96-111, Southco mini offset hinge male 96-112

1. Separate Catchment from Catchment Hinges by removing mounting fasteners from underside of Catchment.
2. Remove Catchment Hinge Mounts from Patient Support by unscrewing mounting screws.
3. Remove Catchment Hinges from Catchment Hinge Mounts by unscrewing mounting screws.
4. Attach new Catchment Hinges on Catchment Hinge Mounts by screwing in mounting screws.
5. Attach Catchment Hinge Mounts to Patient Support by screwing in mounting screws.
6. Hold Catchment in position while fastening Catchment Hinges to Catchment.

10.12 Patient Support Wheel

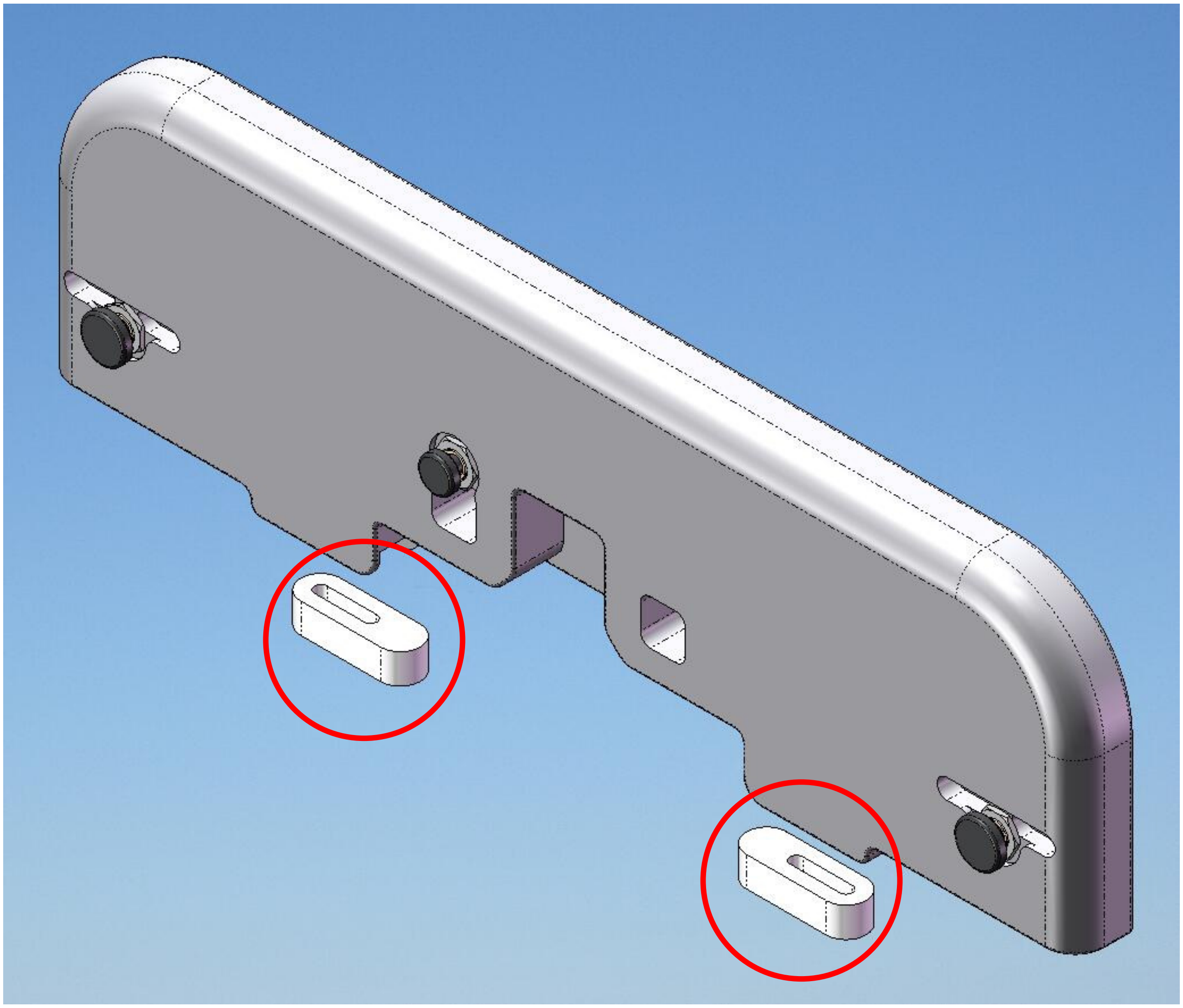
Time Required: 50 minutes
Tools Required: Non-magnetic medium slot screwdriver
Parts Required: AR10B-GL-1/2



1. Access underside of Patient Support.
2. Remove all fasteners wheel mounts adjacent to the wheel required for replacement.
3. Separate wheel mounts from wheels and axles.
4. Inspect wheels and replace as required.
5. Re-assemble wheels on axles between wheel mounts.
6. Replace Wheel Mounts on Patient Support and screw in fasteners.

10.13 L-R Centering Bumper

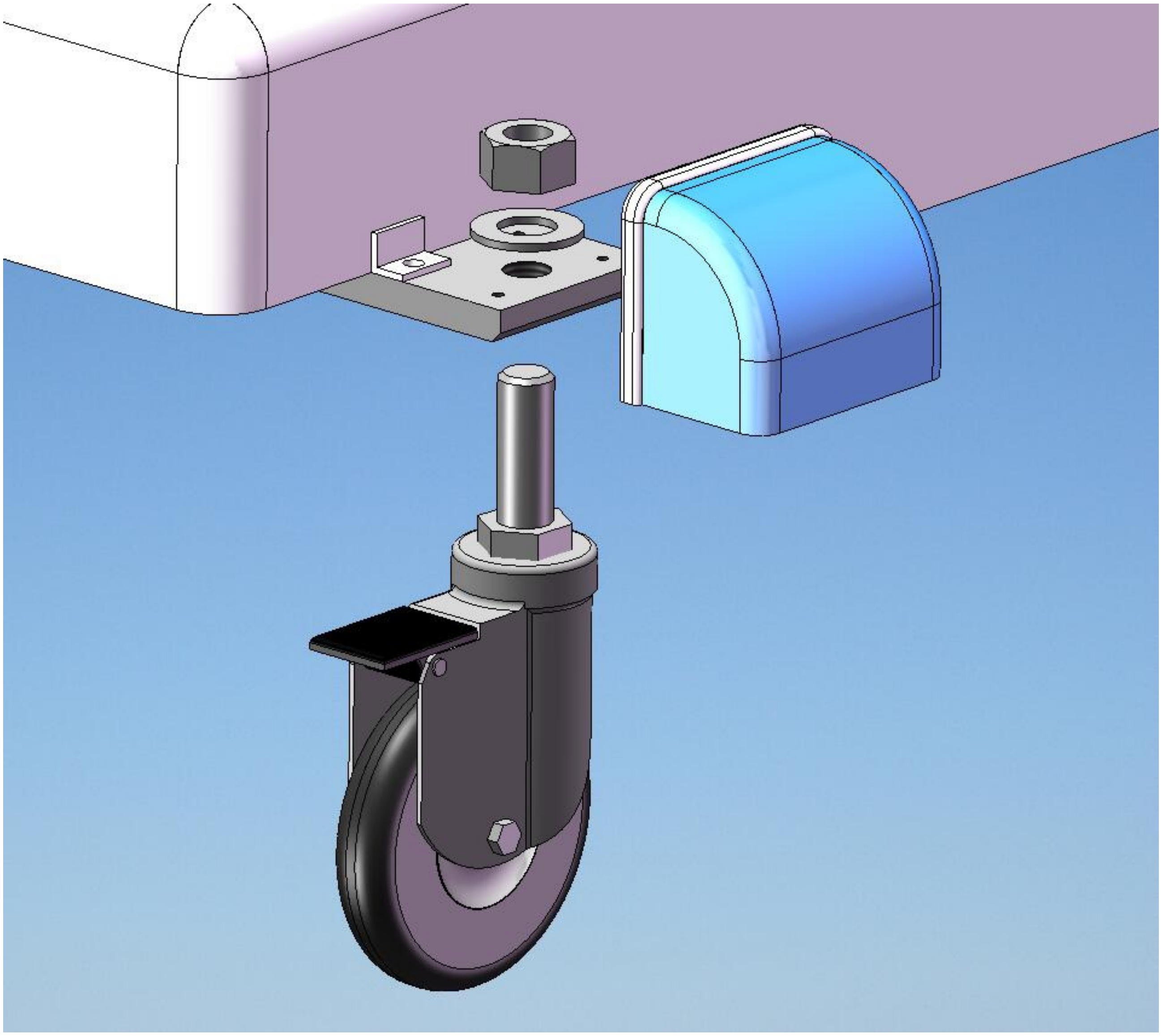
Time Required: 40 minutes
Tools Required: ¼” non-magnetic Allen wrench
Parts Required: BEP-FRM-005-003



1. Note position of Centering Bumper and remove from bottom of Bumper by unscrewing mounting screw.
2. Insert mounting screw in Centering Bumper and screw into Bumper and measure to ensure correct position.

10.14 Caster

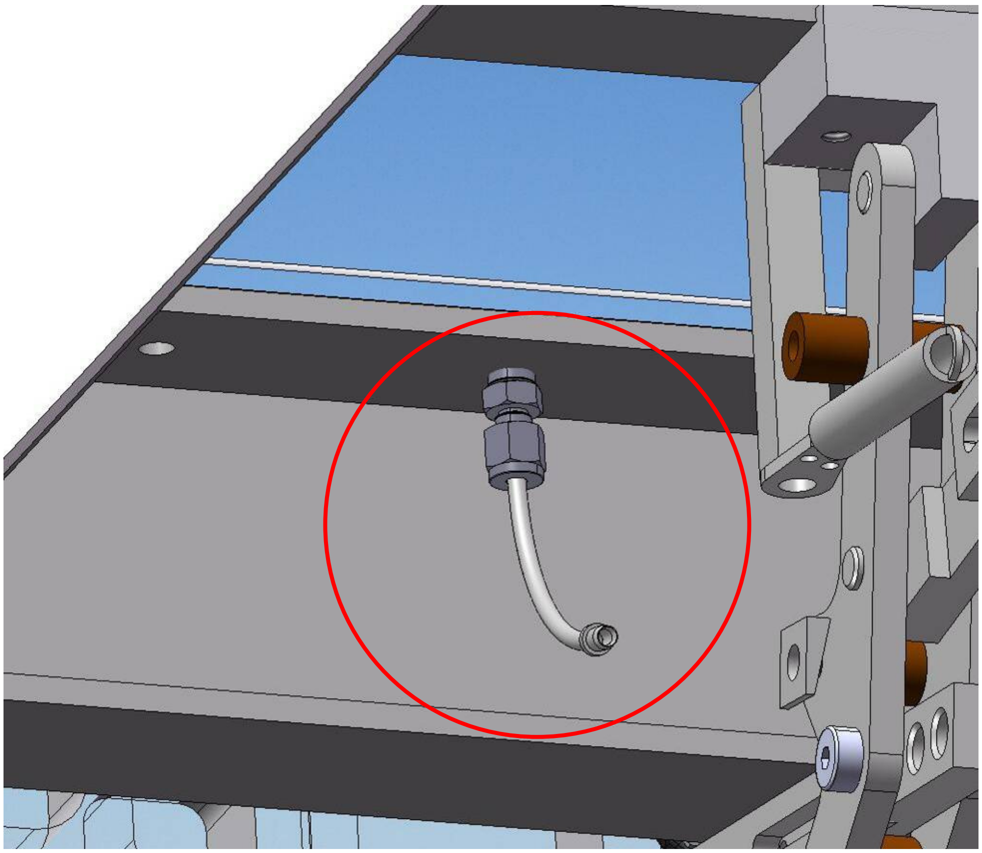
Time Required: 40 minutes
Tools Required: 1 ½” non-magnetic wrench and 5/32” Allen wrench
Parts Required: SSBH-TBL-6510-EL-DEL



1. Lift end of stretcher and place on toolkit or similar sturdy support..
2. Loosen mounting screws and remove the Wheel Covers.
3. Loosen 1.5” locknut
4. Note position of caster’s fine height adjustment by counting turns or thread
5. Unscrew completely from plate.
6. Screw in new Caster to correct position and tighten nut very tight.
7. Replace Wheel Cover and tighten mounting screws.

10.15 Cable Elbow

Time Required: 40 minutes
Tools Required: Non-magnetic adjustable wrench and 5/32” or 3/16” Allen wrench
Parts Required: 5000260-11 or 5000261-11



1. Detach cable from No-undock hook or Swing Assembly and remove cable from cable elbow.
2. Unscrew Cable Elbow Assembly from mounting point with adjustable wrench.
3. Insert cable through new Cable Elbow Assembly and screw into mounting point with adjustable wrench.
4. Fasten cable to No-undock hook or Swing Assembly and refer to section 7.15 or 7.12 to re-set mounting point.

11.0 Vanguard PM Procedure

This procedure should be executed semi-annually for low use systems and quarterly for high use systems. A high use system is one that is used to perform >30 procedures weekly and a low use is one which is used for <30.

Any service activities performed on a system are always logged in the SMI service database. Before servicing a unit in the field, the service technician should confirm the date of last service for that particular unit. If the unit is, or will soon be due for a preventative maintenance call, one should be carried out while the service technician is on site, time permitting.

Before servicing the Vanguard, the SMI service technician should inspect the scanner and its accessories to look for any obvious damage or signs of wear. Any perceived problems should be reported to the GE FE for that particular site. The SMI service technician should also speak with a site staff member who has personally used the system and ask if they have any functionality concerns to report.

Dock the system and advance to scan listen for any noises which could indicate broken wheels, interferences between the patient support and the scanner and stretcher, or within the patient support itself. It is sometimes helpful to clean the scanner's bridge of any debris or dust.

11.1 Verification of Settings

In order to verify the settings of the Vanguard a series of functionality tests must be made. If any of these tests are failed, refer to the troubleshooting section in order to restore full functionality.

- Dock stretcher and take note of docking force required and any noise, vibrations or side-to-side play
- Check doghouse actuator sensor to make sure it is not pushing the pin on the docking fixture to far in and possibly overloading it.
- Check gap between stretcher and scanner to ensure it is 1"
- Advance doghouse to table and ensure doghouse mechanism is properly actuated.
- Advance patient support completely into magnet with weight applied and ensure it is not contacting bore or making any creaking, scratching, scraping, clicking, grinding, vibrating or unusual noises.
- Ensure latch tab is completely retracted into LBS and is not protruding.
- Verify UBS/LBS surfaces are parallel and level with scanner's bridge
- Use tool to ensure that guide rails are collinear.
- Test docking force by pushing stretcher hard toward scanner and by applying moderate left and right force to stretcher handle.
- At the end of the LBS nearest the scanner apply heavy left right force to check LR centering of the stretcher.
- Test patient support emergency release and re-latch patient support.
- Check no-undock hook to ensure it is hanging freely and effectively preventing undock pedal from being pressed when the patient support is advanced into the scanner.
- Return patient support to home position, again checking for unusual noises.
- Check undock hook to ensure it is well clear of undock pedal arm.
- Press undock pedal and leave system against scanner.
- Ensure that the doghouse unlatches and moves into scanner.
- Check docking hook height to make sure it is not contacting scanner dock's bronze hook when rolling the stretcher up to the scanner.
- Perform Phantom Scan Quality Assurance Procedure SMI-0217 and compare results with baseline figures entered from first scan on install.
- Complete all image quality testing procedures issued for this product
- Refer to all service bulletins for this product
- Perform any Service Actions Pending for this product. If you do not have an up to date list of actions outstanding, telephone Sentinelle Service for an update.

11.2 Subsystem Tests and inspection

Stretcher

- Check the rolling and swivel brakes of all four casters
- Actuate travel stop mechanism by hand, inspect cable at mount points
- No dock spring function and appearance.
- Latch tab actuation and height, condition of cable and cable elbows
- Function of bridge, and articulation of latch and stopper
- Check function of lighting system, individual LEDs
- By hand check articulation of the docking hook to ensure that it moves up and down freely without binding.
- Check the undock spring to ensure it is not contacting any other parts of the linkage other than its mounting hardware.

- Apply moderate left and right force to the dock and undock pedals to check if they have become worn and are contacting other components of the stretcher or docking mechanism.
- While actuating the dock pedal by hand look at the drive link to make sure it is not rubbing against the tension coupling or center plate.
- Dock the system and then, while undocking, check to ensure that the undock pedal arm does not contact the link wheel pin.
- Remove drawers and inspect latch tab cable and cable elbows for wear or fatigue.
- Check function of side handles to ensure they extend and retract smoothly and that gap covers are moving freely..
- Check all fasteners on moving parts of docking mechanism for tightness.
- Check drawers for smooth function and integrity of sliders.
- Check all system labels are present and intact.

Patient Support

- Actuate and test emergency release with table in scanner several times
- Check all wheels on the underside of the patient support for excessive noise (needing lubrication or replacement) or damage such as cracks.
- Lift foot end of patient support and hang over end of LBS to expose the doghouse mechanism and check that mechanism functions properly and isn't damaged or worn.
- Ensure the nut on the doghouse mechanism plunger screw is tight using a small wrench.
- Check patient support for cracks or damage to gelcoat or further structural damage.
- Check the headrest for smooth operation and any cracks or damage.
- Check the padding to ensure that is has no rips or damaged Velcro.
- Check all system labels are present and intact.

11.3 RF and Compression System

Inspect all of the coils to ensure that they are not cracked or otherwise damaged.
Check the coil cables for any rips, strains or loose connections.
Inspect the cable and balun at the end of the cable tray for damage.
Inspect the system plug by looking for bent pins and ensuring that it connects smoothly to the MRI with moderate force and good retention.
Check the sliders, frames and compression rails for damage or wear and that the slider locks lock firmly on the compression rail and frames.
Check contralateral support for cracking or damage.
Check all system labels are present and intact.

11.4 Cleaning

Clean all surfaces of the patient support and stretcher with alcohol and a soft cloth.
Make sure there is no tape or tape residue on any of the surfaces that would interfere with normal operation of the stretcher patient support or RF system
Flip over the patient support and inspect all of the wheels on the underside of the patient support and ensure that they are clean, and rolling smoothly and quietly. Brush away any dust or debris and lubricate with Teflon lubricant as required.
Clean the mirrors and catchment with alcohol if necessary.
Check the casters to ensure they have no debris caught in them which could inhibit their ability to roll or swivel.
Clean any dirt or debris from the stretcher drawers.
Check and clean contralateral support.
Clean any of the padding that may require it with Virox wipes.

11.5 Follow-up

Upon completion of the PM, report condition of Vanguard to site manager and leave a completed calling card. If any wear or damage has been detected due to misuse, consider making site manager aware of what has occurred and suggest actions to take to prevent future damage or premature wear to system.